

Matrices

Matrix - A matrix A of order $m \times n$ is arrangement of mn objects in m rows and n columns.

Notation - $A = A_{m \times n} = [a_{ij}]_{m \times n}$

if $a_{ij} \in \mathbb{R}$ (real numbers), A is real matrix.

if $a_{ij} \in \mathbb{C}$ (complex numbers), A is complex matrix.

Row matrix (or row vector) - $[a_1, a_2, \dots, a_n]$.

Column matrix (or column vector) -

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}$$

Transpose of a matrix. $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 1 & -1 & 1 \end{bmatrix}$

$$A' = A^T = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 0 & -1 \\ 3 & 1 & 1 \end{bmatrix}$$

Further, $(A^T)^T = A$.

Note - $(A+B)^T = A^T + B^T$

$$(AB)^T = B^T A^T$$

For matrices $AB \neq BA$ (non-commutative)

for eg - $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Here $\boxed{AB = BA}$

but for $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$B = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}$$

Here $\boxed{AB \neq BA}$.



Symmetric matrix - $A^T = A$

if we see elements $a_{ij}^T = a_{ji} = a_{ij}$

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$$\boxed{a_{ji} = a_{ij}}$$

So upper ~~triangular~~ elements = lower ~~triangular~~ elements
(diagonal)

eg. $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 0 & -1 \\ 3 & -1 & 1 \end{bmatrix}$

skew-symmetric matrix - $A^T = -A$

if we see elements $a_{ij}^T = a_{ji} = -a_{ij}$

if we see diagonal elements ($i=j$)

$$a_{ii} = -a_{ii}$$

$$2a_{ii} = 0$$

$$\boxed{a_{ii} = 0}$$

So all diagonal elements are zero.
and upper ~~triangular~~ (diagonal) elements
= - lower diagonal elements.

eg - $\begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & -1 \\ -3 & 1 & 0 \end{bmatrix}$

Note - Every real matrix can be written uniquely as the sum of symmetric and skew-symmetric matrix.

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

where $\frac{1}{2}(A + A^T)$ is symmetric and $\frac{1}{2}(A - A^T)$ is skew-symmetric

Ex- For $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 4 & -1 \\ 2 & -1 & 1 \end{bmatrix}$

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$$A^T = \begin{bmatrix} 1 & -1 & 2 \\ 2 & 4 & -1 \\ 3 & -1 & 1 \end{bmatrix}$$

$$A + A^T = \begin{bmatrix} 2 & 1 & 4 \\ 1 & 0 & -2 \\ 4 & -2 & 2 \end{bmatrix} \quad \frac{1}{2}(A + A^T) = \begin{bmatrix} 1 & 1/2 & 2 \\ 1/2 & 0 & -1 \\ 2 & -1 & 1 \end{bmatrix}$$

$$A - A^T = \begin{bmatrix} 0 & 3 & 2 \\ -3 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \quad \frac{1}{2}(A - A^T) = \begin{bmatrix} 0 & 3/2 & 1 \\ -3/2 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

Clearly $\frac{1}{2}(A + A^T)$ is symmetric and $\frac{1}{2}(A - A^T)$ is skew-symmetric.

Also $A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{sym}} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{skew-sym.}}$

Hermitian matrix

Ex. Show that if A, B are symmetric then AB is symmetric iff $AB = BA$

Pr. Since A, B are sym. $\therefore A^T = A, B^T = B$

T.P. AB is symm, that is $(AB)^T = AB$

Now $(AB)^T = B^T A^T = BA$

$\therefore (AB)^T = AB$ if we take $BA = AB$

Conversely-

I.P. $AB = BA$

Now $(AB)^T = B^T A^T = BA$

Also, given $(AB)^T = AB$

$\therefore \boxed{BA = AB}$

Hermitian matrix:- Here A is a complex matrix. ④

$$\text{let } A = \begin{bmatrix} 1 & i & 1+i \\ 1-i & 0 & 2+3i \\ 2-3i & 0 & -i \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 1 & -i & 1-i \\ 1+i & 0 & 2-3i \\ 2+3i & 0 & i \end{bmatrix} \quad \left\{ \because \overline{a+ib} = a-ib \right\}$$

$$A^{\odot} = (\bar{A})^T = \begin{bmatrix} 1 & 1+i & 2+3i \\ -i & 0 & 0 \\ 1-i & 2-3i & i \end{bmatrix} \quad \text{Further, } (A^{\odot})^{\odot} = A.$$

A complex matrix A is Hermitian matrix if

$$A^{\odot} = (\bar{A})^T = A$$

if we see elements $(\bar{a}_{ij})^T = \bar{a}_{ji} = a_{ij}$

for diagonal elements ($i=j$)

$\bar{a}_{ii} = -a_{ii}$
 $\therefore a_{ii}$ is either zero
 or purely real.

eg- $\begin{bmatrix} 1 & i & 1+i \\ -i & 0 & -i \\ 1-i & i & -i \end{bmatrix}$

Note- let $z = a+ib$
 $\bar{z} = a-ib$

for $\bar{z} = z$

$$\cancel{a-ib} = \cancel{a+ib}$$

$$2ib = 0$$

$b = 0 = \text{imaginary part.}$

So either $a \neq 0$ or $a = 0$
 (real part)

$\therefore z$ is either purely real
 or zero.

for $\bar{z} = -z$

$$a-ib = -(a+ib)$$

$$\cancel{a-ib} = \cancel{-a-ib}$$

$$2a = 0$$

$a = 0 = \text{real part.}$

So either $b \neq 0$ or $b = 0$
 (imaginary part)

$\therefore z$ is either purely
 imaginary or zero.

skew-Hermitian - A ^{complex} matrix A is skew-Hermitian

if $A^\ominus = (\bar{A})^T = -A$

if we see elements $(\bar{a}_{ij})^T = \bar{a}_{ji} = -a_{ij}$

for diagonal elements ($i=j$)

$$\bar{a}_{ji} = -a_{ii}$$

$\therefore a_{ii}$ is either zero or purely imaginary

eg -
$$\begin{bmatrix} 0 & i & 1+i \\ i & 0 & -i \\ -1+i & -i & -i \end{bmatrix}$$

Ex - Every complex matrix can be written uniquely as sum of Hermitian and skew-Hermitian matrix.

$$A = \underbrace{\frac{1}{2} (A + A^\ominus)}_{\text{Hermitian}} + \underbrace{\frac{1}{2} (A - A^\ominus)}_{\text{skew-Hermitian}}.$$

Proof - This is obvious, we can write A as

$$A = \frac{1}{2} (A + A^\ominus) + \frac{1}{2} (A - A^\ominus)$$

Now, we need to prove that $\frac{1}{2} (A + A^\ominus)$ is Hermitian and $\frac{1}{2} (A - A^\ominus)$ is skew-Hermitian.

that is $\left[\frac{1}{2} (A + A^\ominus) \right]^\ominus = \frac{1}{2} (A + A^\ominus)$

and $\left[\frac{1}{2} (A - A^\ominus) \right]^\ominus = -\frac{1}{2} (A - A^\ominus)$

$$\begin{aligned} \left[\frac{1}{2} (A + A^\ominus) \right]^\ominus &= \frac{1}{2} (A + A^\ominus)^\ominus \\ &= \frac{1}{2} (A^\ominus + (A^\ominus)^\ominus) \\ &= \frac{1}{2} (A^\ominus + A) \\ &= \frac{1}{2} (A + A^\ominus) \end{aligned} \quad \left\{ \begin{array}{l} \because (A^\ominus)^\ominus \\ = (\bar{A}^T)^T \\ = (A^T)^T \\ = A. \end{array} \right.$$

$\therefore \frac{1}{2} (A + A^\ominus)$ is Hermitian.

$$\begin{aligned}
 \left(\frac{1}{2} (A - A^{\odot}) \right)^{\odot} &= \frac{1}{2} (A - A^{\odot})^{\odot} \\
 &= \frac{1}{2} (A^{\odot} - (A^{\odot})^{\odot}) \\
 &= \frac{1}{2} (A^{\odot} - A) \\
 &= -\frac{1}{2} (A - A^{\odot})
 \end{aligned} \tag{6}$$

$\therefore \frac{1}{2} (A - A^{\odot})$ is skew-Hermitian.

Now, we will prove uniqueness.

From above, we see A can be written as sum of Hermitian $\frac{1}{2} (A + A^{\odot})$ and skew-Hermitian $\frac{1}{2} (A - A^{\odot})$ matrices.

We assume A can also be written as

$$A = P + Q$$

where P is Hermitian $P^{\odot} = P$

and Q is skew-Hermitian $Q^{\odot} = -Q$

We need to prove $P = \frac{1}{2} (A + A^{\odot})$
and $Q = \frac{1}{2} (A - A^{\odot})$

pf. According to our assumption

$$A = P + Q$$

$$A^{\odot} = (P + Q)^{\odot} = P^{\odot} + Q^{\odot} = P - Q \quad \leftarrow$$

$$\therefore A + A^{\odot} = 2P$$

$$A - A^{\odot} = 2Q$$

$$\frac{1}{2} (A + A^{\odot}) = P$$

$$\frac{1}{2} (A - A^{\odot}) = Q$$

Hence, A can only be written as

$$A = \underbrace{\frac{1}{2} (A + A^{\odot})}_{\text{Hermitian}} + \underbrace{\frac{1}{2} (A - A^{\odot})}_{\text{skew-Hermitian}}$$

where $\frac{1}{2} (A + A^{\odot})$ is Hermitian

and $\frac{1}{2} (A - A^{\odot})$ is skew-Hermitian.

$$A^{\odot} = \begin{bmatrix} 1-i & 1 & 1+i \\ 2-3i & 0 & 2+3i \\ -i & 0 & -i \end{bmatrix}$$

⑦

$$\frac{1}{2}(A + A^{\odot}) = \frac{1}{2} \begin{bmatrix} 2 & 1+2i & 3+4i \\ 1-2i & 0 & 2+3i \\ 3-4i & 2-3i & 0 \end{bmatrix} \text{ is Hermitian.}$$

$$\text{and } \frac{1}{2}(A - A^{\odot}) = \frac{1}{2} \begin{bmatrix} 0 & -1 & -1-2i \\ 1 & 0 & 2+3i \\ 1-2i & -2+3i & -2i \end{bmatrix} \text{ is skew-Hermitian.}$$

$$\therefore A = \underbrace{\frac{1}{2}(A + A^{\odot})}_{\text{Hermitian}} + \underbrace{\frac{1}{2}(A - A^{\odot})}_{\text{skew-Hermitian}}$$

Unitary matrix - $AA^T = I$ (identity matrix).

Determinant -

$$D = \text{diag}(d_1, d_2, \dots, d_n)$$

$$\det(D) = d_1 d_2 \dots d_n$$

$$\det(AB) = \det A \det B$$

$$\det(-A_{n \times n}) = (-1)^n \det A$$

Hermitian matrix:- Here A is a complex matrix. ④

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 or purely real.

eg- $\begin{bmatrix} 1 & i & 1+i \\ -i & 0 & -i \\ 1-i & i & -i \end{bmatrix}$

Note- let $z = a+ib$
 $\bar{z} = a-ib$

for $\bar{z} = z$

$$\cancel{a-ib} = \cancel{a+ib}$$

$$2ib = 0$$

$$b = 0 = \text{imaginary part.}$$

So either $a \neq 0$ or $a = 0$
 (real part)

$\therefore z$ is either purely real
 or zero.

for $\bar{z} = -z$

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skew-Hermitian - A ^{complex} matrix A is skew-Hermitian

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eg -
$$\begin{bmatrix} 0 & i & 1+i \\ i & 0 & -i \\ -1+i & -i & i \end{bmatrix}$$

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Hence, A can only be written as

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where $\frac{1}{2} (A + A^{\odot})$ is Hermitian

and $\frac{1}{2} (A - A^{\odot})$ is skew-Hermitian.

$$\underline{\text{Ex-}} \quad A = \begin{bmatrix} 1 & i & 1+i \\ 1-i & 0 & 2+3i \\ 2-3i & 0 & -i \end{bmatrix} \quad \textcircled{7}$$

$$A^0 = \begin{bmatrix} 1 & 1+i & 2+3i \\ -i & 0 & 0 \\ 1-i & 2-3i & -i \end{bmatrix}$$

$$\frac{1}{2}(A+A^0) = \frac{1}{2} \begin{bmatrix} 2 & 1+2i & 3+4i \\ 1-2i & 0 & 2+3i \\ 3-4i & 2-3i & 0 \end{bmatrix} \text{ is Hermitian.}$$

$$\text{and } \frac{1}{2}(A-A^0) = \frac{1}{2} \begin{bmatrix} 0 & -1 & -1-2i \\ 1 & 0 & 2+3i \\ 1-2i & -2+3i & -2i \end{bmatrix} \text{ is skew-Hermitian.}$$

$$\therefore A = \underbrace{\frac{1}{2}(A+A^0)}_{\text{Hermitian}} + \underbrace{\frac{1}{2}(A-A^0)}_{\text{skew-Hermitian}}$$

Unitary matrix - $AA^T = I$ (identity matrix).

Determinant -

$$D = \text{diag}(d_1, d_2, \dots, d_n)$$

$$\det(D) = d_1 d_2 \dots d_n$$

$$\det(AB) = \det A \det B$$

$$\det(-A_{n \times n}) = (-1)^n \det A$$

Rank of a matrix - rank of a matrix is the order of the largest non-zero minor of the matrix.
(or it is the order of largest submatrix of the given matrix whose determinant is non-zero).

for $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 0 & 2 & 4 \end{bmatrix}$ so rank = 2.

minors are $\begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \\ 0 & 2 & 4 \end{bmatrix}$ of order 3×3 $\det = 0$
 $\hookrightarrow \det \neq 0$ of order 2×2

$$\begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 3 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 \\ 2 & 4 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 0 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 3 \\ 2 & 4 \end{bmatrix}, \begin{bmatrix} -1 & 3 \\ 0 & 4 \end{bmatrix}, [1], [2], [3], [-1], [0], [1], [0], [2], [2], [4], [3], [4]$$

Now consider submatrices of order $m \times n$

Notation - rank of $A = \rho(A)$

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For a matrix A of order $m \times n$
 $\rho(A) \leq \min(m, n)$.

Note - rank of a null matrix = 0

Ex $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}_{2 \times 3}$

$$\rho(A) \leq \min(2, 3) = 2$$

$$\det \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix} = 1 \neq 0$$

$$\therefore \underline{\rho(A) = 2}$$

Ex $A = \begin{bmatrix} 1 & 3 & -4 \\ -1 & -3 & 4 \\ -2 & 6 & -8 \end{bmatrix}$ Here $\rho(A) \leq 3$

$\det A = 0$ as $R_2 = -R_1$, $R_3 = 2R_1$

$$\rho(A) \neq 3$$

for $\rho(A) < 3 \Rightarrow \rho(A) \leq 2$

we consider all 2×2 submatrices

all have $\det = 0$ ←

So $\rho(A) \neq 2 \Rightarrow \rho(A) < 2$

since A is not a null matrix

$$\therefore \underline{\rho(A) = 1}$$

Ex $A = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & -1 \\ 1 & -1 & 2 \\ 5 & 4 & -5 \end{bmatrix}_{4 \times 3}$

Here $\rho(A) \leq \min(4, 3) = 3$

submatrices of order 3×3

$$\rho(A) < 3$$

$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & -1 \\ 1 & -1 & 2 \end{bmatrix}, \begin{bmatrix} 2 & 1 & -1 \\ 1 & -1 & 2 \\ 5 & 4 & -5 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 \\ 1 & -1 & 2 \\ 5 & 4 & -5 \end{bmatrix}$

$R_1 + R_3 = R_2$ $\begin{bmatrix} 1 & 2 & -3 \\ 2 & 1 & -1 \\ 5 & 4 & -5 \end{bmatrix}$

$\downarrow$
 $\cancel{2R_1} + R_2 = R_3$

We can get $\det = 0$ for all submatrices of order 3×3 .

Now consider submatrices of order 2×2 .

$$\det \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} = -3 \neq 0$$

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$$\therefore \underline{P(A) = 2}$$

System of linear equations -

$$x - y + z = 4$$

$$2x + y - 3z = 0$$

$$x + y + z = 2$$

$$\begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$$

$$AX = b$$

$$\text{where } A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } b = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}.$$

On general,

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

$$\therefore AX = b$$

$$\text{where } A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \text{ and } b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

if $b \neq 0$ then $AX = b$ is non-homogeneous system of equations.

if $b = 0$ then $AX = 0$ is homogeneous system of equations.

Cramer's rule - $AX = b$ ($b \neq 0$)

unique sol if $|A| \neq 0$ $X = A^{-1}b$ } consistent.

infinite sol if $|A| = 0$ and $|A_j| = 0$

no sol if $|A| = 0$ and at least one $|A_j| \neq 0$.

for $AX = 0$

unique sol (trivial sol) $X = 0$ or infinite sol (non trivial) $X \neq 0$.

$$\underline{\text{Ex.}} \quad \begin{bmatrix} 1 & -1 & -3 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix} \quad (12)$$

$$AX = b$$

$$\det A = \begin{vmatrix} 1 & -1 & -3 \\ 2 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = 1(1+3) + 1(2+3) + 1(2-1) \\ = 4 + 5 + 1 = 10 \neq 0$$

$\therefore$ unique sol.

$$X = A^{-1}b = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$$

$$\therefore x = 2, y = -1 \text{ and } z = 1$$

$$\underline{\text{OR.}} \quad |A_1| = \begin{vmatrix} 4 & -1 & -3 \\ 0 & 1 & 1 \\ 2 & 1 & 1 \end{vmatrix} = 20$$

replace 1st column with b

$$|A_2| = \begin{vmatrix} 1 & 4 & -1 \\ 2 & 0 & -3 \\ 1 & 2 & 1 \end{vmatrix} = -10$$

replace 2nd column with b.

$$|A_3| = \begin{vmatrix} 1 & -1 & 4 \\ 2 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix}$$

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$$|A_3| = \begin{vmatrix} 1 & -1 & 4 \\ 2 & 1 & 0 \\ 1 & 1 & 2 \end{vmatrix} = 10$$

replace 3rd column with b

$$\therefore x = \frac{|A_1|}{|A|} = 2, \quad y = \frac{|A_2|}{|A|} = -1 \quad \text{and} \quad z = \frac{|A_3|}{|A|} = 1$$

$$\underline{\text{Ex-}} \quad \begin{bmatrix} 1 & -1 & 3 \\ 2 & 3 & 1 \\ 3 & 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

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$$AX = b$$

$$\text{since } R_1 + R_2 = R_3$$

$$\text{so } |A| = |A_1| = |A_2| = |A_3| = 0$$

$\therefore$ infinite sol.

$$x_1 - x_2 + 3x_3 = 3 \Rightarrow x_1 - x_2 = 3 - 3x_3$$

$$2x_1 + 3x_2 + x_3 = 2 \Rightarrow 2x_1 + 3x_2 = 2 - x_3$$

$$\text{On solving } \begin{array}{r} 3x_1 - 3x_2 = 9 - 9x_3 \\ 2x_1 + 3x_2 = 2 - x_3 \end{array}$$

$$5x_1 = 11 - 10x_3$$

$$x_1 = \frac{11 - 10x_3}{5}$$

$$-2x_1 + 2x_2 = -6 + 6x_3$$

$$2x_1 + 3x_2 = 2 - x_3$$

$$5x_2 = -4 + 5x_3$$

$$x_2 = \frac{-4 + 5x_3}{5} \text{ where } x_3 \text{ is arbitrary.}$$

$$\underline{\text{Ex-}} \quad \begin{bmatrix} 4 & 9 & 3 \\ 2 & 3 & 1 \\ 2 & 6 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 7 \end{bmatrix}$$

$$AX = b$$

$$\text{since } R_1 = R_2 + R_3 \text{ holds for } |A| \therefore |A| = 0$$

$$|A_1| = \begin{vmatrix} 6 & 9 & 3 \\ 2 & 3 & 1 \\ 2 & 6 & 2 \end{vmatrix} = 0$$

$$|A_2| = \begin{vmatrix} 4 & 6 & 3 \\ 2 & 2 & 1 \\ 2 & 7 & 2 \end{vmatrix} = 6 \neq 0$$

$$|A_3| = \begin{vmatrix} 4 & 9 & 6 \\ 2 & 3 & 2 \\ 2 & 6 & 7 \end{vmatrix} = 18 \neq 0$$

$\therefore$ no sol.

Hence, the system of equations is inconsistent.

$$\underline{\text{Ex 6}} \quad \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & -2 \\ 4 & 7 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (12)$$

$$AX = 0$$

Since $2R_1 + R_2 = R_3 \quad \therefore |A| = 0$

so ^{non} trivial (infinite) sol.

$$x + 2y = -3z$$

$$2x + 3y = 2z$$

$$-y = 8z$$

$$y = -8z$$

$x = 13z$ where z is arbitrary.

$$\underline{\text{Ex 6}} \quad \begin{bmatrix} 1 & 2 & 1 \\ 1 & 4 & 3 \\ 1 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$AX = 0$$

$$\text{Here } |A| = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 4 & 3 \\ 1 & 4 & 1 \end{vmatrix} = 1(4-12) - 2(1-3) + 1(4-4) \\ = -8 + 4 = -4 \neq 0$$

$\therefore$ trivial sol. $X = A^{-1}0 = 0$.

$$\Rightarrow \underline{x = y = z = 0.}$$

Trace of a matrix = sum of diagonal elements of a matrix.

if $A^k = O$ for smallest k
then A is called nilpotent matrix with nilpotency k .
if $A^2 = I$ then A is called idempotent matrix.

Since $| \alpha A | = \alpha^n |A|$

$\therefore | -A | = (-1)^n |A|$

Inverse of a matrix - $A^{-1} = \frac{\text{Adj } A}{|A|}$

if $D = \text{diag}(d_1, d_2, \dots, d_n)$

then $D^{-1} = \text{diag}(1/d_1, 1/d_2, \dots, 1/d_n)$

Ex- Find rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ Here $p \leq 3$

since $R_1 + R_2 = R_3$ $\det = 0$ $p < 3$

Also $\det \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} = 0 \neq 0$ $\therefore p = 2$

Ex- $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ Here $p \leq 4$

$\sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 2 & 3 & -1 & -1 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ $R_1 \rightarrow R_2$

$\sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 5 & 7 & 7 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$ $R_2 - 2R_1$
 $R_3 - 3R_1$
 $R_4 - 6R_1$

$\sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & 3/5 & 7/5 \\ 0 & 4 & 9 & 10 \\ 0 & 9 & 12 & 17 \end{bmatrix}$ $R_2 \rightarrow R_2 / 5$

$$\sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & 3/5 & 7/5 \\ 0 & 0 & 33/5 & 29/5 \\ 0 & 0 & 33/5 & 29/5 \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow 4R_2 \\ R_4 - 9R_2 \end{array}$$

$$\sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & 3/5 & 7/5 \\ 0 & 0 & 33/5 & 29/5 \\ 0 & 0 & 0 & 20/5 \end{bmatrix} \quad R_4 \rightarrow R_3 / 3$$

$$\sim \begin{bmatrix} 1 & -1 & -2 & -4 \\ 0 & 1 & 1 & 7/5 \\ 0 & 0 & 1 & -79/25 \\ 0 & 0 & 0 & \frac{344}{25} \end{bmatrix} \quad R_4 - 3R_3$$

$$\det = \frac{344}{25} \neq 0 \quad \therefore p = 4$$

$$\det = 0 \quad p \neq 4 \quad p < 4$$

$$\det \begin{pmatrix} 1 & -1 & -2 \\ 0 & 1 & 3/5 \\ 0 & 0 & 33/5 \end{pmatrix} = \frac{33}{5} \neq 0 \quad \underline{p = 3}$$

ex- Find μ such that $p(A) = 3$

$$\text{where } A = \begin{bmatrix} \mu & -1 & 0 & 0 \\ 0 & \mu & -1 & 0 \\ 0 & 0 & \mu & -1 \\ -6 & 11 & -6 & 1 \end{bmatrix}$$

Since order of A is $4 \times 4 \quad \therefore p \leq 4$

But $p = 3 < 4 \quad \therefore \det(A) = 0$

$$\Rightarrow \mu^3 - 6\mu^2 + 11\mu - 6 = 0$$

$$\Rightarrow (\mu - 1)(\mu - 2)(\mu - 3) = 0$$

$$\therefore \mu = 1, 2, 3$$

$$\det \begin{pmatrix} \mu & -1 & 0 \\ 0 & \mu & -1 \\ 0 & 0 & \mu \end{pmatrix} = \mu^3 \neq 0 \quad \text{for } \mu = 1, 2, 3$$

$$\therefore p(A) = 3 \quad \text{for } \underline{\mu = 1, 2, 3}$$

Ex. Find λ such that the system of equations
 $x - \lambda y + z = 0$, $\lambda x + 3y - \lambda z = 0$,
 $3x + y - z = 0$ has
 (i) only trivial sol and (ii) non-trivial sol.

We have $AX = 0$

$$\begin{bmatrix} 1 & -\lambda & 1 \\ \lambda & 3 & -\lambda \\ 3 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore$ if $|A| \neq 0$ we have trivial sol.

$$\begin{vmatrix} 1 & -\lambda & 1 \\ \lambda & 3 & -\lambda \\ 3 & 1 & -1 \end{vmatrix} \neq 0$$

$$\Rightarrow 1(-3 + \lambda) + \lambda(-\lambda + 3\lambda) + 1(-\lambda + 9\lambda) \neq 0$$

$$\Rightarrow -3 + \lambda + 2\lambda^2 + 8\lambda \neq 0$$

$$\Rightarrow 2\lambda^2 + 9\lambda - 3 \neq 0$$

$$\Rightarrow \lambda^2 + \lambda - 3 \neq 0$$

$$\Rightarrow (\lambda + 3)(\lambda - 2) \neq 0$$

$$\therefore \lambda \neq -3, 2$$

Also if $|A| = 0$ we have non-trivial sol.

$$\therefore |A| = 0$$

$$\Rightarrow \lambda = -3, 2$$

Ex. Find λ and μ such that the system of equations
 $x + 2y + z = 6$, $x + 4y + 3z = 10$,
 $x + 4y + \lambda z = \mu$

has (i) a unique sol,
 (ii) infinite sol. and
 (iii) no sol.

We have $AX = b$

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 4 & 3 \\ 1 & 4 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix}$$

For unique sol $|A| \neq 0$

$$\begin{vmatrix} 1 & 2 & 1 \\ 1 & 4 & 3 \\ 1 & 4 & \lambda \end{vmatrix} \neq 0$$

$$\Rightarrow 1(4\lambda - 12) - 2(\lambda - 3) + 1(4 - 4) \neq 0$$

$$\Rightarrow 4\lambda - 12 - 2\lambda + 6 \neq 0$$

$$\Rightarrow 2\lambda - 6 \neq 0$$

$$\Rightarrow \lambda \neq 3 \text{ and } \mu \text{ is arbitrary.}$$

for infinite sol

$$|A| = |A_1| = |A_2| = |A_3| = 0$$

$$|A| = 0 \Rightarrow \lambda = 3$$

$$|A_1| = \begin{vmatrix} 6 & 9 & 1 \\ \mu & 4 & 3 \end{vmatrix} = 0$$

$$|A_2| = \begin{vmatrix} 1 & 6 & 1 \\ 1 & \mu & 3 \end{vmatrix} = 0$$

$$|A_3| = \begin{vmatrix} 1 & 9 & 6 \\ 1 & 4 & \mu \end{vmatrix} = 0$$

$$\Rightarrow \mu = 10.$$

for no sol.

$$|A| = 0 \text{ and at least one of } |A_i| \neq 0$$

$$\therefore \lambda = 3 \text{ and } \mu \neq 10.$$

Ex. Find rank of $A = \begin{bmatrix} 1 & 1 & 1 \\ p & q & r \\ p^3 & q^3 & r^3 \end{bmatrix}$

$$|A| = 1(qr^3 - r^3q) - 1(pr^3 - p^3r) + 1(pq^3 - p^3q)$$

$$= qr(r^2 - q^2) - pr(r^2 - p^2) + pq(q^2 - p^2)$$

$$= qr(r+q)(r-q) - pr(r+p)(r-p) + pq(q+p)(q-p)$$

$$+ pq(q+p)(q-p)$$

$$= -(p-q)(q-r)(r-p)(p+q+r)$$

$$\rho(A) = 3 \text{ if } \det \neq 0$$

$$-(p-q)(q-r)(r-p)(p+q+r) \neq 0$$

$$p \neq q, q \neq r, r \neq p, p+q+r \neq 0$$

$$\rho(A) = 2 \text{ if } \det = 0$$

$$p = q \text{ or } q = r \text{ or } r = p \text{ or } p+q+r = 0$$

$$\therefore p = q \neq r$$

$$\rho(A) = 1 \text{ if } \det \text{ of all submatrices of order } 2 \times 2 = 0$$

$$\Rightarrow p = q = r.$$

Echelon Form of a Matrix -

The space generated by rows of matrix A , is called row-space of A and $\dim(\text{row space of } A) = \text{row rank of } A$. (14)

row echelon form if no. of zeros preceeding the first non-zero entry of a row increases row by row untill a row having all zero entries is obtained.

eg- $\begin{bmatrix} 1 & 3 & 5 & 7 \\ 0 & 5 & 4 & 1 \\ 0 & 0 & 0 & 9 \end{bmatrix}, \begin{bmatrix} 1 & -1 & 2 & 3 \\ 0 & 0 & 3 & 5 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

no. of non-zero rows in row echelon form of $A = \text{rank of } A$ and these rows forms basis of row space of A .

Note-(i) For square matrix, row echelon form is upper triangular matrix and column echelon form is lower triangular matrix.

(ii) For list of vectors, if row rank = order of matrix $\Rightarrow$ list.

①. Reduce -

(a) $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 5 \\ 1 & 5 & 5 & 7 \\ 8 & 1 & 14 & 17 \end{bmatrix}$ to row echelon form

$R_2 - 2R_1, R_3 - R_1, R_4 - 8R_1$
 $= \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -2 & -3 \\ 0 & 3 & 2 & 3 \\ 0 & -15 & -10 & -15 \end{bmatrix}$ $R_3 + R_2, R_4 - 5R_2 = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\therefore \rho(A) = 2$.

(b) to column echelon form.

$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 5 \\ 1 & 5 & 5 & 7 \\ 8 & 1 & 14 & 17 \end{bmatrix}$ $C_2 - 2C_1; C_3 - 3C_1; C_4 - 4C_1$

$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & -3 & -2 & -3 \\ 1 & 3 & 2 & 3 \\ 8 & -15 & -10 & -15 \end{bmatrix}$ $R_3 - \frac{3}{2}C_2; C_4 - C_2$

$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & -3 & -2 & -3 \\ 1 & 3 & 2 & 3 \\ 8 & -15 & -10 & -15 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & -3 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 8 & -15 & 0 & 0 \end{bmatrix} \therefore \underline{\rho(A) = 2}$

(2) l.i. or l.d.

(15)

(a) $(1,1,0,1); (1,1,1,1); (-1,1,1,1); (1,0,0,1)$

$$A = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 2 \\ 0 & -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$$

$R_2 - R_1, R_3 + R_1, R_4 - R_1 \quad R_2 \rightarrow R_3 \quad 2R_4 + R_2$

$$= \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 2 & 1 & 2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$$

$R_4 - R_3 \quad \therefore \rho(A) = 4 \Rightarrow \text{l.i. and forms basis.}$

(b) $(2,2,0,2); (4,1,4,1); (3,0,4,0)$

$$A = \begin{bmatrix} 2 & 2 & 0 & 2 \\ 4 & 1 & 4 & 1 \\ 3 & 0 & 4 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 0 & 2 \\ 0 & -3 & 4 & -3 \\ 0 & -6 & 8 & -6 \end{bmatrix} = \begin{bmatrix} 2 & 2 & 0 & 2 \\ 0 & -3 & 4 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$R_2 - 2R_1, 2R_3 - 3R_2 \quad R_3 - 2R_2$

$\therefore \rho(A) = 2 \Rightarrow \text{l.d. and forms basis.}$

Gauss-Elimination Method for Non-homogeneous System-

$A_{m \times n} X_{n \times 1} = b_{m \times 1}$ i.e. m eq's in n - unknowns

let $r = \text{row rank of } A$.

(a) $r < m$; $\rho(A) < \rho(A|b) \Rightarrow \text{no sol.}$

(b) $m \geq n$; $r = n$ $\rho(A) = \rho(A|b) = n \Rightarrow \text{unique sol.}$

(c) $r < n$ $\rho(A) = \rho(A|b)$ if $m < n$ and $\rho(A) = \rho(A|b) \Rightarrow \text{infinite sol.}$

① Find sol. of the following eq's by using Gauss

Elimination - Method -

(a) $\begin{bmatrix} 2 & -3 & 1 \\ 1 & -1 & 2 \\ 2 & 1 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix}$

$$[A|b] = \begin{bmatrix} 2 & -3 & 1 & -2 \\ 1 & -1 & 2 & 3 \\ 2 & 1 & -3 & -2 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 & 3 \\ 2 & -3 & 1 & -2 \\ 2 & 1 & -3 & -2 \end{bmatrix}$$

$R_1 \rightarrow R_2 \quad R_2 - 2R_1 \quad R_3 - 2R_1$

$$= \begin{bmatrix} 1 & -1 & 2 & | & 3 \\ 0 & -1 & -3 & | & -8 \\ 0 & 3 & -7 & | & -8 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 2 & | & 3 \\ 0 & -1 & -3 & | & -8 \\ 0 & 0 & -16 & | & -32 \end{bmatrix}$$

$$R_3 + 3R_2$$

$$\therefore \rho(A) = \rho(A|b) = 3 = n$$

$\Rightarrow$ unique sol.

$$-16z = -32$$

$$z = 2$$

$$-y - 3z = -8$$

$$-y = -8 + 6$$

$$y = 2$$

$$x - y + 2z = 3$$

$$x = 3 - y + 2z$$

$$x = 3$$

$$(6) \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\therefore [A|b] = \begin{bmatrix} 1 & 4 & 7 & | & 1 \\ 2 & 5 & 8 & | & 2 \\ 1 & 2 & 3 & | & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 7 & | & 1 \\ 0 & -3 & -6 & | & 0 \\ 0 & -2 & -4 & | & 0 \end{bmatrix}$$

$$R_2 - 2R_1; R_3 - R_1 \quad 3R_3 + 2R_2$$

$$= \begin{bmatrix} 1 & 4 & 7 & | & 1 \\ 0 & -3 & -6 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$\therefore \rho(A) = 2 < 3 = n$$

$\Rightarrow$ infinite sol.

$$x + 4y + 2z = 1$$

$$-3y - 2z = 0$$

$$y = -2z$$

$$x = 1 + 2z$$

let $z = \alpha$, (α arbitrary)

$$\therefore x = 1 + 2\alpha$$

$$y = -2\alpha$$

$$z = \alpha$$

$$(c) \begin{bmatrix} 1 & -4 & 7 \\ 3 & 8 & -2 \\ 7 & -8 & +26 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \\ 3 \end{bmatrix}$$

$$\therefore [A|b] = \begin{bmatrix} 1 & -4 & 7 & | & 8 \\ 3 & 8 & -2 & | & 6 \\ 7 & -8 & +26 & | & 3 \end{bmatrix} = \begin{bmatrix} 1 & -4 & 7 & | & 8 \\ 0 & 20 & -23 & | & -18 \\ 0 & 20 & -23 & | & -53 \end{bmatrix}$$

$$R_2 - 3R_1; R_3 - 7R_1$$

$$R_3 - R_2$$

$$= \begin{bmatrix} 1 & -4 & 7 & | & 8 \\ 0 & 20 & -23 & | & -18 \\ 0 & 0 & 0 & | & -35 \end{bmatrix}$$

$$\therefore \rho(A) = 2 < 3 = \rho(A|b)$$

$\Rightarrow$ no solution.

$$(2) \begin{bmatrix} 1 & -2 & 1 & 2 \\ 1 & 1 & -1 & 1 \\ 1 & 7 & -5 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \quad (16)$$

$$\therefore [A|b] = \left[\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 1 & 1 & -1 & 1 & 2 \\ 1 & 7 & -5 & -1 & 4 \end{array} \right] = \left[\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 0 & 3 & -2 & -1 & 1 \\ 0 & 9 & -6 & -3 & 3 \end{array} \right]$$

$$\begin{array}{l} R_2 - R_1; R_3 - R_1 \\ R_3 - 3R_2 \end{array}$$

$$= \left[\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 0 & 3 & -2 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \therefore \rho(A) = \rho(A|b) = 2 < 4 = n$$

$\Rightarrow$ infinite sol.

$$x - 2y + z + 2w = 1$$

$$3y - 2z - w = 1$$

let $z = \alpha, w = \beta$ (α, β arbitrary)

$$\therefore x = 1 + 2 + 4\alpha + 2\beta - \alpha - 2\beta = 3 + \alpha - 4\beta$$

$$y = \frac{1 + 2\alpha + \beta}{3}, z = \alpha, w = \beta$$

Note-

For Homogenous system of eqⁿs - $AX = 0$

if $\rho(A) = n \Rightarrow$ trivial sol.

and if $\rho(A) < n \Rightarrow$ infinite sol.

sol. space = null space with dim = nullity of A.

$$\therefore \rho(A) + n(A) = n$$

$$(3)(a) \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ -1 & 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore [A|b] = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ -1 & 1 & 1 & -1 & 0 \\ 1 & 1 & -1 & 1 & 0 \\ -1 & 1 & -1 & 1 & 0 \end{array} \right] = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 2 & 2 & 2 & 0 \\ 0 & 0 & 2 & 2 & 0 \\ 0 & 0 & -2 & 0 & 0 \end{array} \right] = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 2 & 2 & 2 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & -2 & 0 & 0 \end{array} \right]$$

$$R_2 + R_1, R_3 + R_1, R_4 - R_1, R_3 + R_4$$

$$R_3 \rightarrow R_4 = \left[\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 0 \\ 0 & 2 & 2 & 2 & 0 \\ 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \end{array} \right] \therefore \rho(A) = 4 = n$$

$\Rightarrow$ trivial sol.

i.e. $x = y = z = w = 0$

$$(b) \begin{bmatrix} 1 & 1 & -2 & -1 \\ 2 & 1 & 1 & -2 \\ 3 & 2 & -1 & -3 \\ 4 & 2 & 2 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\therefore A = \begin{bmatrix} 1 & 1 & -2 & -1 \\ 2 & 1 & 1 & -2 \\ 3 & 2 & -1 & -3 \\ 4 & 2 & 2 & -4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & -2 & -1 \\ 0 & -1 & 5 & 0 \\ 0 & -1 & 5 & 0 \\ 0 & -2 & 10 & 0 \end{bmatrix} \begin{array}{l} R_3 - R_2; \\ R_4 - 2R_2; \\ R_2 - 2R_1, R_3 - 3R_1, R_4 - 4R_1 \end{array}$$

$$= \begin{bmatrix} 1 & 1 & -2 & -1 \\ 0 & -1 & 5 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \therefore \rho(A) = 2 < 4 = n$$

$\Rightarrow$ infinite sol.

$$\begin{array}{l} x + y - 2z - w = 0 \\ -y + 5z = 0 \end{array} \quad \begin{array}{l} \text{let } z = \alpha, w = \beta; \\ (\alpha, \beta \text{ arbitrary}) \end{array}$$

$$\therefore x = \alpha - 3\beta$$

$$y = 5\beta, z = \alpha, w = \beta$$

Gauss-Jordan Method to find the inverse of a Matrix -

$$\textcircled{4} A = \begin{bmatrix} -1 & 1 & 2 \\ 3 & -1 & 1 \\ -1 & 3 & 4 \end{bmatrix} \quad \text{by } (-1)R_1$$

$$\therefore [A|I] = \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 0 & 0 \\ 3 & -1 & 1 & 0 & 1 & 0 \\ -1 & 3 & 4 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} R_2 + 3R_1; \\ R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 2 & 7 & 3 & 1 & 0 \\ 0 & 2 & 2 & -1 & 0 & 1 \end{array} \right] \quad R_2/7$$

$$= \left[\begin{array}{ccc|ccc} 1 & -1 & -2 & 1 & 0 & 0 \\ 0 & 1 & 7/2 & 3/2 & 1/2 & 0 \\ 0 & 2 & 2 & -1 & 0 & 1 \end{array} \right] \begin{array}{l} R_1 + R_2 \\ R_3 - 2R_2 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 3/2 & 1/2 & 1/2 & 0 \\ 0 & 1 & 7/2 & 3/2 & 1/2 & 0 \\ 0 & 0 & -5 & -4 & -1 & 1 \end{array} \right] \quad -R_3/5$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 3/2 & 1/2 & 1/2 & 0 \\ 0 & 1 & 7/2 & 3/2 & 1/2 & 0 \\ 0 & 0 & 1 & 4/5 & 1/5 & -1/5 \end{array} \right] \begin{array}{l} R_1 - 3R_3/2; \\ R_2 - 7R_3/2 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 13/10 & 2/10 & 3/10 \\ 0 & 1 & 0 & 13/10 & -2/10 & 7/10 \\ 0 & 0 & 1 & 4/5 & 1/5 & -1/5 \end{array} \right]$$

$$\therefore A^{-1} = \frac{1}{10} \begin{bmatrix} 13 & 2 & 3 \\ -13 & -2 & 7 \\ 8 & 2 & -2 \end{bmatrix}$$

Ex-3.42 Find the eigenvalues and the corresponding eigenvectors of the following matrices -

$$(i) A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^3 = 0$$
$$\lambda = 1, 1, 1$$

for $\lambda = 1$

$$(A - I)X = 0$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_2 = 0$$

$$x_3 = 0$$

Since we have no equation involving x_1 , so take x_1 arbitrary

$$\therefore X = \begin{pmatrix} x_1 \\ 0 \\ 0 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

eigenvector for $\lambda = 1$ is $X_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

$$(ii) A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^3 = 0$$
$$\lambda = 1, 1, 1$$

for $\lambda = 1$

$$(A - I)X = 0$$

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_2 = 0$$

Since we have no equation involving x_1 and x_3 .
So take x_1 and x_3 arbitrary.

$$X = \begin{pmatrix} x_1 \\ 0 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ x_3 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

∴ eigenvectors for $\lambda = 1$ are

$$X_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \text{ and } X_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

$$(iii) A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^3 = 0$$
$$\lambda = 1, 1, 1$$

for $\lambda = 1$

$$(A - I)X = 0$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Since we have no equation involving x_1, x_2 and x_3 so take x_1, x_2 and x_3 arbitrary.

$$X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ x_2 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ x_3 \end{pmatrix}$$
$$= x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

∴ eigenvectors for $\lambda = 1$ are

$$X_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, X_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \text{ and } X_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

$$(4) A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 2 \\ 0 & 2-\lambda & 1 \\ -1 & 2 & 2-\lambda \end{vmatrix}$$

$$= (1-\lambda)(4 + \lambda^2 - 4\lambda - 2) - 2(0 + 1) + 2(0 + 2 - \lambda)$$

$$= (1-\lambda)(\lambda^2 - 4\lambda + 2) - 2 + 4 - 2\lambda$$

$$= (1-\lambda)(\lambda^2 - 4\lambda + 2) + 2(1-\lambda)$$

$$= (1-\lambda)(\lambda^2 - 4\lambda + 4)$$

$$= (1-\lambda)(2-\lambda)^2 = 0$$

$$\lambda = 1, 2, 2.$$

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

$$(A - \lambda I) = \begin{bmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{bmatrix}$$

$$= (-2-\lambda)(\lambda^2 - \lambda - 12) - 2(-2\lambda - 6) - 3(4 + 1 - \lambda)$$

$$= (-2-\lambda)(\lambda^2 - \lambda - 12) + 4\lambda + 12 + 3\lambda$$

$$= -2\lambda^2 + 2\lambda + 24 - \lambda^3 + \lambda^2 + 12\lambda + 17\lambda + 31$$

$$= \lambda^3 + \lambda^2 + 21\lambda + 45 = 0$$

for $\lambda = -3$

$$27 - 9 - 63 + 45 = 0$$

To find remaining roots

$$\begin{array}{r|rrrr} -3 & -1 & -1 & 21 & 45 \\ & & 3 & -6 & -45 \\ \hline & -1 & +2 & 15 & 0 \end{array}$$

$$-\lambda^2 + 2\lambda + 15 = 0$$

$$(\lambda - 5)(\lambda + 3) = 0$$

$$\lambda = 5, -3$$

So eigenvalues are $\lambda = 5, -3, -3$

for $\lambda = 5$

$$(A - 5I)X = 0$$

$$\begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -7x_1 + 2x_2 - 3x_3 = 0 \\ 2x_1 - 4x_2 - 6x_3 = 0 \\ -x_1 - 2x_2 - 5x_3 = 0 \end{bmatrix} \rightarrow \begin{bmatrix} -8x_1 - 8x_3 = 0 \\ x_1 = -x_3 \end{bmatrix}$$

$$\begin{bmatrix} 2x_1 - 4x_2 - 6x_3 = 0 \\ -2x_1 - 4x_2 - 10x_3 = 0 \end{bmatrix}$$

$$\begin{bmatrix} 4x_1 + 4x_3 = 0 \\ x_1 = -x_3 \end{bmatrix}$$

for $\lambda = 1$

$$(A - I)X = 0$$

$$\begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{aligned} 2x_2 + 2x_3 &= 0 \\ x_2 + x_3 &= 0 \end{aligned} \rightarrow \begin{aligned} x_2 + x_3 &= 0 \\ \boxed{x_2 = -x_3} \end{aligned}$$

$$-x_1 + 2x_2 + x_3 = 0$$

$$-x_1 - 2x_3 + x_3 = 0$$

$$-x_1 - x_3 = 0$$

$$\boxed{x_1 = -x_3}$$

$$\therefore X = \begin{pmatrix} -x_3 \\ -x_3 \\ x_3 \end{pmatrix} = -x_3 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$\therefore$ eigenvector for $\lambda = 1$ is $X_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$.

for $\lambda = 2$

$$(A - 2I)X = 0$$

$$\begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x_1 + 2x_2 + 2x_3 = 0$$

$$x_3 = 0$$

$$-x_1 + 2x_2 = 0$$

$$x_1 = 2x_2$$

$$\therefore X = \begin{pmatrix} 2x_2 \\ x_2 \\ 0 \end{pmatrix} = x_2 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

$\therefore$ eigenvector for $\lambda = 2$ is $X_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$.

$$X_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$$

if we put $x_3 = 0$ it becomes

$$-x_1 + 2x_2 = 0$$

so first and third equations are identical.

$$-x_1 - 2x_2 - 5x_3 = 0$$

$$-x_1 - 2x_2 - 5(-x_3) = 0$$

$$4x_1 - 2x_2 = 0$$

$$x_2 = 2x_1$$

$$\therefore X = \begin{pmatrix} x_1 \\ 2x_1 \\ -x_1 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$$

$\therefore$ eigenvector for $\lambda = -5$ is $X_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$.

for $\lambda = -3$

$$(A + 3I)X = 0$$

$$\begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 + 2x_2 - 3x_3 = 0$$

$$2x_1 + 4x_2 - 6x_3 = 0$$

$$-x_1 - 2x_2 + 3x_3 = 0$$

all equations are same

$$x_1 + 2x_2 - 3x_3 = 0$$

$$x_1 = -2x_2 + 3x_3$$

$$X = \begin{pmatrix} -2x_2 + 3x_3 \\ x_2 \\ x_3 \end{pmatrix} = x_2 \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$$

$\therefore$ eigenvectors for $\lambda = -3$ are

$$X_2 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} \text{ and } X_3 = \begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}.$$

Ex-3-46. The eigenvectors for a 3×3 matrix A corresponding to the eigenvalues 1, 1, 3 are $(1, 0, -1)^T$, $(0, 1, -1)^T$ and $(1, 1, 0)^T$ respectively. Find the matrix A .

$$\text{let } A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$\text{for } \lambda = 1 \quad X_1 = (1, 0, -1)^T$$

$$AX = \lambda X$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$$

$$a_{11} - a_{13} = 1 \rightarrow \textcircled{1}$$

$$a_{21} - a_{23} = 0 \rightarrow \textcircled{2}$$

$$a_{31} - a_{33} = -1 \rightarrow \textcircled{3}$$

$$\text{for } \lambda = 1 \quad X_1 = (0, 1, -1)^T$$

$$AX = \lambda X$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} = 1 \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

$$a_{12} - a_{13} = 0 \rightarrow \textcircled{4}$$

$$a_{22} - a_{23} = 1 \rightarrow \textcircled{5}$$

$$a_{32} - a_{33} = -1 \rightarrow \textcircled{6}$$

$$\text{for } \lambda = 3 \quad X_3 = (1, 1, 0)^T$$

$$AX = \lambda X$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$a_{11} + a_{12} = 3 \rightarrow \textcircled{7}$$

$$a_{21} + a_{22} = 3 \rightarrow \textcircled{8}$$

$$a_{31} + a_{32} = 0 \rightarrow \textcircled{9}$$

$$\textcircled{4} - \textcircled{7} \Rightarrow -a_{11} - a_{13} = -3$$

$$\textcircled{1} \Rightarrow a_{11} - a_{13} = 1$$

$$\begin{array}{r} -2a_{13} = -2 \Rightarrow a_{13} = 1 \end{array}$$

$$\text{Put in } \textcircled{1} \text{ and } \textcircled{4} \text{ implies } a_{11} = 2, a_{12} = 1$$

$$\textcircled{5} - \textcircled{8} \Rightarrow -a_{21} - a_{23} = 0 \Rightarrow a_{21} = a_{23}$$

$$\text{Put in } \textcircled{5} \text{ and } \textcircled{8} \text{ implies } a_{22} - a_{21} = 1$$

$$a_{21} + a_{22} = 3$$

$$\begin{array}{r} 2a_{22} = 2 \end{array}$$

$$\textcircled{2} - \textcircled{5} \Rightarrow \begin{array}{r} a_{21} - a_{23} = 0 \\ -a_{22} + a_{23} = 1 \end{array}$$

$$a_{21} - a_{22} = -1$$

$$\textcircled{8} \Rightarrow a_{21} + a_{22} = 3$$

$$\begin{array}{r} -2a_{22} = -4 \\ a_{22} = 2 \end{array}$$

$$a_{22} = 2$$

$$a_{21} - 2 = -1$$

$$a_{21} = 1$$

$$a_{21} = a_{23} = 1$$

$$\textcircled{3} - \textcircled{6} \Rightarrow$$

$$a_{31} - a_{32} = 0$$

$$a_{31} = a_{32}$$

$$\textcircled{4} \Rightarrow$$

$$a_{31} + a_{32} = 0$$

$$a_{31} = a_{32} = 0$$

$$a_{31} - a_{33} = -1$$

$$a_{33} = 1$$

$$\therefore A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$



Eigen values and Eigen Vectors -

(17)

$$AX = \lambda X \Rightarrow (A - \lambda I)X = 0$$

non-trivial sol if $|A - \lambda I| = 0$

$$\Rightarrow \text{characteristic polynomial in } \lambda = 0 \Rightarrow \lambda^n - c_1 \lambda^{n-1} + c_2 \lambda^{n-2} - \dots + (-1)^{n-1} c_{n-1} \lambda + (-1)^n c_n = 0$$

Note - ① $\sum \lambda = \text{trace}(A)$ and $\prod \lambda = |A|$

② For a real matrix A , if $\alpha + i\beta$ is its eigen value then $\alpha - i\beta$ must also be its eigen value.

Cayley - Hamilton Theorem - Every square matrix A satisfies its own characteristic equation, that is -

$$A^n - c_1 A^{n-1} + \dots + (-1)^{n-1} c_{n-1} A + (-1)^n c_n I = 0$$

Proof - $|A - \lambda I| = \text{polynomial in } \lambda \text{ with degree } \leq n$

$\therefore \text{adj}(A - \lambda I) = \text{polynomial in } \lambda \text{ with degree } \leq n-1$

$$\Rightarrow \text{adj}(A - \lambda I) = B_1 \lambda^{n-1} + B_2 \lambda^{n-2} + \dots + B_{n-1} \lambda + B_n$$

$$\therefore (A - \lambda I) \text{adj}(A - \lambda I) = |A - \lambda I| I$$

$$\therefore (A - \lambda I) (B_1 \lambda^{n-1} + B_2 \lambda^{n-2} + \dots + B_{n-1} \lambda + B_n) = |A - \lambda I| I$$

$$= \lambda^n I - c_1 \lambda^{n-1} I + c_2 \lambda^{n-2} I + \dots + (-1)^{n-1} c_{n-1} I + (-1)^n c_n I$$

$$\lambda^n \quad -B_1 = I$$

$$\lambda^{n-1} \quad AB_1 - B_2 = -c_1 I$$

$$\lambda^{n-2} \quad AB_2 - B_3 = c_2 I$$

...

$$\lambda^{n-n} \quad AB_{n-1} - B_n = (-1)^{n-1} c_{n-1} I$$

$$\text{constant term} \quad AB_n = (-1)^n c_n I$$

Premultiplying by $A^n, A^{n-1}, \dots, A, I$ respectively and adding; we get -

$$A^n - c_1 A^{n-1} + \dots + (-1)^{n-1} c_{n-1} A + (-1)^n c_n I = 0.$$

which proves the theorem.

(h)

① Verify Cayley-Hamilton theorem for the matrix - (18)

$$A = \begin{bmatrix} -1 & 2 & 0 \\ -1 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Also (i) obtain A^{-1} and A^3 ; (ii) find eigenvalues of A, A^2 and verify that eigenvalues of A^2 are squares of those of A ; (iii) find the spectral radius of A .

Sol- $|A - \lambda I| = -\lambda^3 + 3\lambda^2 - \lambda + 3 = 0$

$$-A^3 + 3A^2 - A + 3I = 0 \Rightarrow \text{verified.}$$

$$(i) A^{-1} = \frac{1}{3} (A^2 - 3A + I) = \frac{1}{3} \begin{bmatrix} -3 & -2 & 4 \\ 3 & 1 & -2 \\ -3 & 0 & 3 \end{bmatrix}$$

$$A^3 = 3A^2 - A + 3I = \begin{bmatrix} -1 & 10 & 12 \\ 1 & 11 & 10 \\ -1 & 16 & 17 \end{bmatrix}$$

(ii) For A $\lambda = 3, i, -i$

For A^2 $|A^2 - \lambda I| = (\lambda + 1)(\lambda + 9)(\lambda + 1) = 0$
 $\therefore \lambda = 9, -1, -1$

(iii) $\rho(A) = \max(\lambda_i) = 3$.

② If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$; then show that -

$A^n = A^{n-2} + A^2 - I$ for $n \geq 3$. Hence find A^{50} .

Sol- $|A - \lambda I| = \lambda^3 - \lambda^2 - \lambda + 1 = 0$

$$\therefore A^3 - A^2 - A + I = 0$$

$$A^3 - A^2 = A - I$$

Premultiplying both sides successively by A ; we obtain -

$$A^3 - A^2 = A - I$$

$$A^4 - A^3 = A^2 - A$$

$$A^{n-1} - A^{n-2} = A^{n-3} - A^{n-4}$$

$$A^n - A^{n-1} = A^{n-2} - A^{n-3}$$

On adding; we get -

$$A^n - A^2 = A^{n-2} - I \Rightarrow A^n = A^{n-2} + A^2 - I \text{ for } n \geq 3.$$

(h)

$$\begin{aligned}
 A^n &= (A^{n-4} + A^2 - I) + A^2 - I = A^{n-4} + 2(A^2 - I) \\
 &= (A^{n-6} + A^2 - I) + 2(A^2 - I) \\
 &= A^{n-6} + 3(A^2 - I) \\
 &= A^{n-(n-2)} + \frac{1}{2}(n-2)(A^2 - I) \\
 &= A^2 + \frac{1}{2}(n-2)(A^2 - I) \\
 &= \frac{n}{2}A^2 - \frac{1}{2}(n-2)I
 \end{aligned}$$

For $n=50$

$$A^{50} = 25A^2 - 24I = \begin{bmatrix} 25 & 0 & 0 \\ 0 & 25 & 0 \\ 0 & 0 & 25 \end{bmatrix}$$

③ Find eigen values and corresponding eigenvectors for -

(a) $A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$

$$|A - \lambda I| = \lambda^2 - 2\lambda + 2 = 0 \Rightarrow \lambda = 1 \pm i$$

For $\lambda = 1 + i$

$$(A - \lambda I)X = 0$$

$$\begin{bmatrix} -i & 1 \\ -1 & -i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} -ix_1 + x_2 = 0 \\ -x_1 - ix_2 = 0 \end{cases}$$

$$\rightarrow -x_1 - ix_2 = 0$$

$$x_2 = i \Rightarrow x_1 = -i$$

$$X = \begin{bmatrix} -i & 1 \end{bmatrix}^T$$

For $\lambda = 1 - i$

$$\begin{bmatrix} i & 1 \\ -1 & i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{cases} ix_1 + x_2 = 0 \\ -x_1 + ix_2 = 0 \end{cases}$$

$$\rightarrow -x_1 + ix_2 = 0$$

$$x_2 = i \Rightarrow x_1 = i$$

$$\therefore X = \begin{bmatrix} i & 1 \end{bmatrix}^T$$

$$(b) A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 2 & 0 & 3 \end{bmatrix}$$

(19)

$$|A - \lambda I| = (1-\lambda)(2-\lambda)(3-\lambda) = 0 \quad \therefore \lambda = 1, 2, 3$$

For $\lambda = 1$

$$(A - I)X = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_2 + x_3 = 0 \quad \Rightarrow x_2 = -x_3$$

$$x_1 + x_3 = 0 \quad \Rightarrow x_1 = -x_3$$

$$\therefore X = [-1, -1, 1]^T$$

For $\lambda = 2$

$$(A - 2I)X = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 0 & 0 \\ 2 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = 0, x_3 = 0, x_2 = \text{arbitrary}$$

$$\therefore X = [0, 1, 0]^T$$

For $\lambda = 3$

$$(A - 3I)X = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -1 & 0 \\ 2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_1 = 0, -x_2 + x_3 = 0 \Rightarrow x_2 = x_3$$

$$\therefore X = [0, 1, 1]^T$$

Note- if all λ_i are distinct $\Rightarrow$ eigenvectors are l.i.

if λ_i are of multiplicity n and $P(A - \lambda I) = r$ then $n - r = \text{l.i. eigen vectors}$.

$$(c) A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 2 \\ 0 & 2-\lambda & 1 \\ -1 & 2 & 2-\lambda \end{vmatrix} = (1-\lambda)(4+\lambda^2-4\lambda-2)$$

$$= (1-\lambda)(\lambda^2-4\lambda+2) - 2(1) + 2(2-\lambda)$$

$$= \lambda^2 - 4\lambda + 2 - \lambda^3 + 4\lambda^2 - 2\lambda + 2 - 2\lambda$$

$$= -\lambda^3 + 5\lambda^2 - 8\lambda + 4 = 0$$

Put $\lambda = 1$

$$-1 + 5 - 8 + 4 = 0$$

$$D = P^{-1}AP \dots$$

$$\begin{array}{c|ccc} 1 & -1 & 5 & -8 & 4 \\ & & -1 & 4 & 4 \\ \hline & -1 & 4 & -4 & 0 \end{array}$$

$$-\lambda^2 + 4\lambda - 4 = 0$$

$$\lambda = 1, 2, 2$$

$$\text{For } \lambda = 1 \quad (A - \lambda I)X = \begin{bmatrix} 0 & 2 & 2 \\ 0 & 1 & 1 \\ -1 & 2 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$x_2 + x_3 = 0$$

$$-x_1 + 2x_2 + x_3 = 0$$

$$x_3 = -1, x_2 = 1, x_1 = 1 \quad \therefore X = [1 \ 1 \ -1]^T$$

$$\text{For } \lambda = 2 \quad (A - 2I)X = \begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-x_1 + 2x_2 + x_3 = 0$$

$$x_3 = 0$$

$$-x_1 + 2x_2 = 0$$

$$\Rightarrow x_1 = 2x_2$$

$$X = [2 \ 1 \ 0]^T$$

Similar matrices - A and B (sq. matrices of some order) are similar if $\exists$ an invertible matrix P $\ni$ $A = P^{-1}BP \Rightarrow PA = BP \Rightarrow PAP^{-1} = B$ (20)

Diagonalizable matrices - if A is similar to a diagonal matrix D i.e. $P^{-1}AP = D$

Note - A sq. matrix A of order n is diagonalizable if and only if it has n linearly independent eigenvectors.

① Show that the matrix $A = \begin{bmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$ is diagonalizable. Hence find P such that $P^{-1}AP$ is a diagonal matrix. Then obtain the matrix B $= A^2 + 5A + 3I$.

Sol - $|A - \lambda I| = \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0 \Rightarrow \lambda = 1, 2, 3$
 $\therefore$ 3 distinct eigenvalues $\Rightarrow$ 3 lin eigenvectors
 $\therefore$ diagonalizable.

$$\text{For } \lambda = 1 \quad (A - I)X = \begin{bmatrix} 2 & 1 & -1 \\ -2 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
$$\therefore X^1 = [1 \quad -1 \quad 1]^T$$

$$\text{For } \lambda = 2 \quad (A - 2I)X = \begin{bmatrix} 1 & 1 & -1 \\ -2 & -1 & 2 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
$$\therefore X^2 = [1 \quad 0 \quad 1]^T$$

$$\text{For } \lambda = 3 \quad (A - 3I)X = \begin{bmatrix} 0 & 1 & -1 \\ -2 & -2 & 2 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
$$\therefore X^3 = [0 \quad 1 \quad 1]^T$$

$$\therefore P = [X^1 \ X^2 \ X^3] = \begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} -1 & -1 & 1 \\ 2 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$P^{-1}AP = \text{diag}(1, 2, 3)$$

$$D = P^{-1}AP \Rightarrow D^2 = P^{-1}A^2P = \text{diag}(1, 4, 9)$$

$\therefore A^2 + 5A + 3I = P(D^2 + 5D + 3I)P^{-1}$
 $= \begin{bmatrix} 25 & 8 & -8 \\ -18 & 9 & 18 \\ -2 & 8 & 19 \end{bmatrix}$

(a) $A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 2 & 1 \\ -1 & 2 & 2 \end{bmatrix}$

$|A - \lambda I| = (1 - \lambda)(2 - \lambda)(2 - \lambda) = 0 \Rightarrow \lambda = 1, 2, 2$

For $\lambda = 2$ $(A - 2I)X = \begin{bmatrix} -1 & 2 & 2 \\ 0 & 0 & 1 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\rho = 2 \Rightarrow n = 3 - 2 = 1$ l.i. eigenvector.

For $\lambda = 1 \Rightarrow 1$ l.i. eigenvector.

$\therefore$ total 2 l.i. eigenvectors

$\Rightarrow$ not diagonalizable.

(b) $A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$

$|A - \lambda I| = \lambda^3 + \lambda^2 - 21\lambda + 45 = 0 \Rightarrow \lambda = 5, -3, -3$

For $\lambda = 5$ $(A - 5I)X = \begin{bmatrix} -7 & 2 & -3 \\ 2 & -4 & -6 \\ -1 & -2 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$X^1 = [1, 2, -1]^T$

For $\lambda = -3$ $(A + 3I)X = \begin{bmatrix} 1 & 2 & -3 \\ 2 & 4 & -6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$x_1 + 2x_2 - 3x_3 = 0$

$\rho = 1 \Rightarrow n = 3 - 1 = 2$ l.i. eigenvectors

$X^2 = [-2, 1, 0]^T$ and $X^3 = [3, 0, 1]^T$

$\therefore$ total 3 l.i. eigenvectors $\Rightarrow$ diagonalizable

$\therefore P = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$ and $P^{-1} = \frac{1}{8} \begin{bmatrix} 1 & 2 & -3 \\ -2 & 4 & 6 \\ 1 & 2 & 5 \end{bmatrix}$

Verify that - $P^{-1}AP = \text{diag}(5, -3, -3)$.

③ Find $A \ni \lambda = 1, 1, 3$ with corresponding $X^1 = (1, 0, -1)^T$, $X^2 = (0, 1, -1)^T$ and $X^3 = (1, 1, 0)^T$ respectively.

Sol- $P = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & -1 & 0 \end{bmatrix}$ and $D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

(21)

$$\therefore P^{-1} = \frac{1}{2} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\therefore A = P^{-1} P = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

④ $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 3 & 1-\lambda & 0 \\ -2 & 0 & 1-\lambda \end{vmatrix} = (1-\lambda)^3 - 6(1-\lambda) + 6(1-\lambda) = 0$$

$$= (1-\lambda)(\lambda^2 + 1 - 2\lambda - 12) = 0$$

$$\Rightarrow \lambda = 1, 1, 1$$

For $\lambda = 1$

$$(A - I)X = \begin{bmatrix} 0 & 2 & 3 \\ 3 & 0 & 0 \\ -2 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\rho = 2 \Rightarrow 3 - 2 = 1$ lii eigenvector.

i.e. $X = [0, 3, -2]^T$

$\Rightarrow$ not diagonalizable.

⑤ Find $P \ni P^{-1}AP = \text{diagonal matrix}$.

$$A = \begin{bmatrix} 1 & 2 & -2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$$

Sol- $|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 & -2 \\ 1 & 1-\lambda & 1 \\ 1 & 3 & -1-\lambda \end{vmatrix} = (1-\lambda)[-(1-\lambda^2) - 3] - 2(-1-\lambda-1) - 2(3-1+\lambda)$

$$= (1-\lambda)(\lambda^2 - 4) - 2(-2-\lambda) - 2(2+\lambda)$$

$$= \lambda^2 - 4 - \lambda^3 + 4\lambda = 0$$

Put $\lambda = 1$ $1 - 4 - 1 + 4 = 0$

$$-\lambda^2 + 4 = 0$$

$$\lambda = \pm 2$$

$$\begin{array}{c|ccc} 1 & -1 & 1 & 4 & -4 \\ & & -1 & 0 & 4 \\ \hline & -1 & 0 & 4 & 0 \end{array}$$

$\Rightarrow$ 3 lii eigenvectors

$\Rightarrow$ diagonalizable.

$$\text{For } \lambda = 1 \quad (A - I)X = \begin{bmatrix} 0 & 2 & -2 \\ 1 & 0 & 1 \\ 1 & 3 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$X^1 = (1, -1, -1)^T$$

$$\text{For } \lambda = 2 \quad (A - 2I)X = \begin{bmatrix} -1 & 2 & -2 \\ 1 & -1 & 1 \\ 1 & 3 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$X^2 = (0, 1, 1)^T$$

$$\text{For } \lambda = -2 \quad (A + 2I)X = \begin{bmatrix} 3 & 2 & -2 \\ 1 & 3 & 1 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$X^3 = (8, -5, 7)^T$$

$$P = \begin{bmatrix} 1 & 0 & 8 \\ -1 & 1 & -5 \\ -1 & 1 & 7 \end{bmatrix}, \quad P^{-1} = \frac{1}{12} \begin{bmatrix} 12 & 8 & -8 \\ 12 & 15 & -3 \\ 0 & -1 & 1 \end{bmatrix}$$

Verify that $P^{-1}AP = \text{diag}(1, 2, -2)$.

1) The eigenvalues of -

(a) an Hermitian matrix are real.

PP- $AX = \lambda X$
 $\bar{X}^T A X = \lambda \bar{X}^T X \Rightarrow \lambda = \frac{\bar{X}^T A X}{\bar{X}^T X}$ - real and positive

$\therefore A$ is hermitian $\therefore \bar{A} = A^T$
 $(\bar{X}^T A X) = X^T \bar{A} \bar{X} = X^T A^T \bar{X} = (X^T A^T \bar{X})^T$
 $= \bar{X}^T A X$
 $\therefore$ real.
 $(\because \bar{X}^T \bar{A} X \text{ is a scalar})$

(b) a skew-hermitian matrix are zero or purely imaginary.

PP- $A^T = -\bar{A}$
 $(\bar{X}^T A X) = \bar{X}^T \bar{A} \bar{X} = -X^T A^T \bar{X} = -(X^T A^T \bar{X})^T$
 $= -(\bar{X}^T A X)$
 $\therefore$ zero or purely imaginary.

(c) an unitary matrix are of magnitude 1.

PP- $A^{-1} = (\bar{A})^T$
 $AX = \lambda X \Rightarrow \bar{A} \bar{X} = \bar{\lambda} \bar{X}$
 $(\bar{A} \bar{X})^T = \bar{X}^T \bar{A}^T = \bar{X}^T \bar{X}^T$
 $\bar{X}^T A^{-1} = \bar{X}^T \bar{X}^T$

$(\bar{X}^T A^{-1})(AX) = \bar{\lambda} \bar{X}^T (\lambda X)$
 $\bar{X}^T X = \bar{\lambda} \lambda \bar{X}^T X$
 $|\lambda|^2 = 1 \Rightarrow |\lambda| = 1 \Rightarrow \lambda = \pm 1.$

T.P. $\lambda = 0$ is a characteristic root of a matrix.
if and only if the matrix is singular.

$$\Rightarrow \text{For } \lambda = 0 \quad |A - \lambda I| = |A| = 0$$

$$\Leftarrow |A - \lambda I| = 0 \Rightarrow |A| - \lambda |I| = 0 \\ \Rightarrow \lambda = 0.$$

T.P. If λ is an eigen value of an orthogonal matrix, then $\frac{1}{\lambda}$ is also its eigen value.

$$\text{as } AA^T = I \Rightarrow A^T = A^{-1}$$

T.P. If A is unitary then A^T is also unitary.

$$AA^* = I = A^*A$$

$$(AA^*)^T = I = (A^*A)^T$$

$$(A^T)^* A^T = I = A^T (A^T)^*.$$

T.P. If A is a unitary matrix then A^{-1} is also unitary.

$$(AA^*)^{-1} = I = (A^*A)^{-1}$$

$$(A^{-1})^* A^{-1} = I = A^{-1} (A^{-1})^*.$$

T.P. A, B unitary $\Rightarrow AB$ unitary.

$$A A^* = I = A^* A \quad \text{and} \quad B B^* = I = B^* B$$

$$\therefore (AB)(AB)^* = A B B^* A^* = A A^* = I$$

$$\text{and } (AB)^*(AB) = B^* A^* A B = B^* B = I.$$

T.P. $\sum \lambda_i = \text{trace of } A$

$\prod \lambda_i = \det A$

let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$\therefore |A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}$

$= -\lambda^3 + \lambda^2(a_{11} + a_{22} + a_{33}) + \dots$

let $\lambda_1, \lambda_2, \lambda_3$ be the eigenvalues of A . then

$|A - \lambda I| = (-1)^3 (\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)$
 $= (-1)^3 \lambda^3 + \lambda^2(\lambda_1 + \lambda_2 + \lambda_3) + \dots$

$\therefore \sum \lambda = \sum a_{ii} = \text{trace of } A$

if $\lambda = 0$ then $|A| = \lambda_1 \lambda_2 \lambda_3 = \prod \lambda_i$

T.P. $1/\lambda$ is eigen value of A^{-1} with same eigen vector

$AX = \lambda X$

$A^{-1}AX = \lambda A^{-1}X$

$\frac{1}{\lambda}X = A^{-1}X = A^T X$ if $AA^T = I$.

T.P. Eigen values of A and A^T are same.

$AX = \lambda X$

$AA^T = I$

$X = \lambda A^T X$ $A^T = A^{-1}$

$A^T Y = \lambda' Y$

Since $(A - \lambda I)^T = A^T - \lambda I^T = A^T - \lambda I$

$\therefore |A - \lambda I| = |A^T - \lambda I|$

So characteristic eqs of A and A^T are same.

$\Rightarrow$ eigen values are same.

Also for distinct eigen values, eigenvectors corresponding to A and A^T are mutually orthogonal.

let $AX = \lambda X$ and $A^T Y = \mu Y$ with $\lambda \neq \mu$.

$Y^T AX = \lambda Y^T X$ and $Y^T A = \mu Y^T$

$\Rightarrow \mu Y^T X = \lambda Y^T X \Rightarrow (\lambda - \mu) Y^T X = 0 \Rightarrow Y^T X = 0$

Q Show that the function

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

is not continuous at $(0, 0)$ but its partial derivatives f_x and f_y exist at $(0, 0)$.

$$\begin{aligned} \lim_{(x, y) \rightarrow (0, 0)} f(x, y) &= \lim_{(x, y) \rightarrow (0, 0)} \frac{xy}{x^2 + y^2} \\ &= \lim_{x \rightarrow 0} \frac{x \cos \theta \cdot x \sin \theta}{x^2 \cos^2 \theta + y^2 \sin^2 \theta} \\ &= \frac{\cos \theta \sin \theta}{\cos^2 \theta + \sin^2 \theta} \end{aligned}$$

$$\left\{ \begin{array}{l} \text{let } x = r \cos \theta \\ y = r \sin \theta \\ x^2 + y^2 = r^2 \\ \text{as } x, y \rightarrow 0 \\ r \rightarrow 0 \\ \text{and } \theta \text{ is arbitrary} \\ (0 \leq \theta \leq 2\pi) \end{array} \right.$$

which is in terms of θ and θ is arbitrary, so limit is not unique and hence limit does not exist.

$\therefore f(x, y)$ is not continuous at $(0, 0)$.

$$\begin{aligned} f_x(0, 0) &= \lim_{\Delta x \rightarrow 0} \frac{f(0 + \Delta x, 0) - f(0, 0)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\frac{(\Delta x)^2 + 0}{(\Delta x)^2 + 0} - 0}{\Delta x} = 0. \end{aligned}$$

$$\begin{aligned} f_y(0, 0) &= \lim_{\Delta y \rightarrow 0} \frac{f(0, 0 + \Delta y) - f(0, 0)}{\Delta y} \\ &= \lim_{\Delta y \rightarrow 0} \frac{\frac{0 + (\Delta y)^2}{0 + (\Delta y)^2} - 0}{\Delta y} = 0. \end{aligned}$$

$\therefore f_x$ and f_y exist at $(0, 0)$.

Q Show that the function

$$f(x, y) = \begin{cases} \frac{x^3 + 2y^3}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

(i) is continuous at $(0, 0)$.

(ii) possesses partial derivatives $f_x(0, 0)$ and $f_y(0, 0)$.

(iii) is not differentiable at $(0, 0)$.

$$\begin{aligned} \lim_{(x, y) \rightarrow (0, 0)} f(x, y) &= \lim_{(x, y) \rightarrow (0, 0)} \frac{x^3 + 2y^3}{x^2 + y^2} \\ &= \lim_{x \rightarrow 0} \frac{x^3 \cos^3 \theta + 2x^3 \sin^3 \theta}{x^2 \cos^2 \theta + x^2 \sin^2 \theta} \\ &= \lim_{x \rightarrow 0} x (\cos^3 \theta + 2 \sin^3 \theta) \\ &= 0 = f(0, 0). \end{aligned}$$

$$\left\{ \begin{array}{l} \text{let } x = r \cos \theta \\ y = r \sin \theta \\ x^2 + y^2 = r^2 \\ \text{as } x, y \rightarrow 0 \\ r \rightarrow 0 \\ \text{and } \theta \text{ is arbitrary} \\ (0 \leq \theta \leq 2\pi) \end{array} \right.$$

$\therefore \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0)$
 $\Rightarrow f(x, y)$ is continuous at $(0, 0)$.

$$f_x(0,0) = \lim_{\Delta x \rightarrow 0} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\frac{(\Delta x)^3}{(\Delta x)^2} - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} = 1$$

$$f_y(0,0) = \lim_{\Delta y \rightarrow 0} \frac{f(0,0+\Delta y) - f(0,0)}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\frac{2(\Delta y)^3}{(\Delta y)^2} - 0}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{2\Delta y}{\Delta y} = 2$$

For $z = f(x, y)$ at (x, y)

$$\Delta z = f(x+\Delta x, y+\Delta y) - f(x, y)$$

Total differential

$$dz = \frac{\partial f}{\partial x} \Delta x + \frac{\partial f}{\partial y} \Delta y$$

$$\Delta p = \sqrt{(\Delta x)^2 + (\Delta y)^2}$$

$z = f(x, y)$ is said to be differentiable at (x, y) if

$$\Delta z = dz + \epsilon_1 \Delta x + \epsilon_2 \Delta y$$

where $\epsilon_1, \epsilon_2 \rightarrow 0$ as $\Delta x, \Delta y \rightarrow 0$.

$$\frac{\Delta z - dz}{\Delta p} = \epsilon_1 \frac{\Delta x}{((\Delta x)^2 + (\Delta y)^2)^{1/2}} + \epsilon_2 \frac{\Delta y}{((\Delta x)^2 + (\Delta y)^2)^{1/2}}$$

$$\lim_{\Delta p \rightarrow 0} \frac{\Delta z - dz}{\Delta p} = \lim_{\Delta x, \Delta y \rightarrow 0} \left\{ \epsilon_1 \frac{\Delta x}{((\Delta x)^2 + (\Delta y)^2)^{1/2}} + \epsilon_2 \frac{\Delta y}{((\Delta x)^2 + (\Delta y)^2)^{1/2}} \right\}$$

$$\text{let } \Delta x = r \cos \theta, \Delta y = r \sin \theta$$

$$(\Delta x)^2 + (\Delta y)^2 = r^2$$

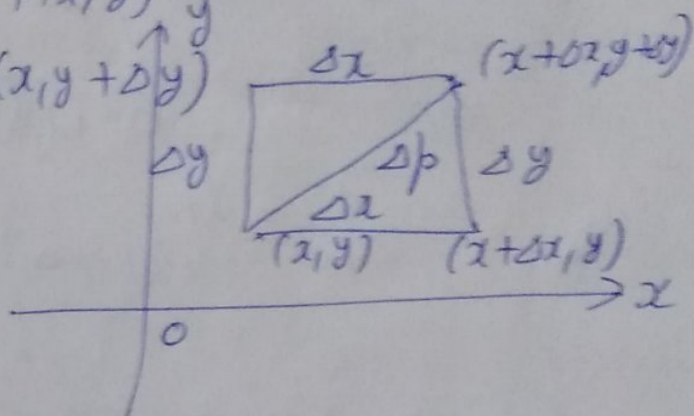
as $\Delta x, \Delta y \rightarrow 0$ $r \rightarrow 0$ and θ is arbitrary $(0 \leq \theta \leq 2\pi)$

Also since $\epsilon_1, \epsilon_2 \rightarrow 0$ as $\Delta x, \Delta y \rightarrow 0$

So $\epsilon_1, \epsilon_2 \rightarrow 0$ as $r \rightarrow 0$

$$\therefore \lim_{\Delta p \rightarrow 0} \frac{\Delta z - dz}{\Delta p} = \lim_{r \rightarrow 0} \left\{ \epsilon_1 \frac{r \cos \theta}{r} + \epsilon_2 \frac{r \sin \theta}{r} \right\}$$

$$= \lim_{r \rightarrow 0} \epsilon_1$$



$$\Rightarrow \lim_{\Delta p \rightarrow 0} \frac{\Delta z - dz}{\Delta p} = 0 \quad (\text{Test for differentiability})$$

$$\therefore \Delta z \equiv dz \quad \text{for } \Delta p \rightarrow 0$$

$$\Delta z \equiv dz \quad \text{for } \Delta x, \Delta y \rightarrow 0$$

Now, at $(0,0)$

$$\Delta z = f(0+\Delta x, 0+\Delta y) - f(0,0)$$

$$= \frac{(\Delta x)^3 + 2(\Delta y)^3}{(\Delta x)^2 + (\Delta y)^2} - 0 = \frac{(\Delta x)^3 + 2(\Delta y)^3}{(\Delta x)^2 + (\Delta y)^2}$$

$$dz = f_x(0,0) \Delta x + f_y(0,0) \Delta y = 1 \cdot \Delta x + 2 \cdot \Delta y$$

$$\therefore \frac{\Delta z - dz}{\Delta p} = \frac{1}{\{(\Delta x)^2 + (\Delta y)^2\}^{1/2}} \left(\frac{(\Delta x)^3 + 2(\Delta y)^3}{(\Delta x)^2 + (\Delta y)^2} - (\Delta x + 2\Delta y) \right)$$

$$\lim_{\Delta p \rightarrow 0} \frac{\Delta z - dz}{\Delta p} = \lim_{\Delta x, \Delta y \rightarrow 0} \left(\frac{(\Delta x)^3 + 2(\Delta y)^3 - (\Delta x + 2\Delta y)((\Delta x)^2 + (\Delta y)^2)}{\{(\Delta x)^2 + (\Delta y)^2\}^{3/2}} \right)$$

$$\text{let } \Delta x = r \cos \theta, \Delta y = r \sin \theta$$

$$\therefore (\Delta x)^2 + (\Delta y)^2 = r^2$$

as $\Delta x, \Delta y \rightarrow 0$, $r \rightarrow 0$ and θ is arbitrary
($0 \leq \theta \leq 2\pi$)

$$\lim_{\Delta p \rightarrow 0} \frac{\Delta z - dz}{\Delta p} = \lim_{r \rightarrow 0} \left(\frac{r^3 \cos^3 \theta + 2r^3 \sin^3 \theta - (r \cos \theta + 2r \sin \theta)(r^2)}{r^3} \right)$$

$$\lim_{\Delta p \rightarrow 0} \frac{\Delta z - dz}{\Delta p} = (\cos^3 \theta + 2 \sin^3 \theta) - (\cos \theta + 2 \sin \theta)$$

which is not unique as it depends upon θ
and θ is arbitrary so limit varies as θ varies.
Thus, limit does not exist

$$\text{and hence } \lim_{\Delta p \rightarrow 0} \frac{\Delta z - dz}{\Delta p} \neq 0$$

So $f(x,y)$ is not differentiable at $(0,0)$.

Partial Derivatives

Functions of one variable $y = f(x)$

eg - $y = x^2$, $y = mx + c$, $y = \sin x$
 $y = \sin^{-1} x$, $y^2 = 4ax \Rightarrow x = \frac{y^2}{4a}$
 $x = g(y)$

Functions of two variables $z = f(x, y)$

eg - $z = \sqrt{1-x^2-y^2}$, $z = \frac{1}{x^2-y^2}$, $z = \log(x+y)$

Functions of three variables $w = f(x, y, z)$

eg - $w = \sqrt{x^2+y^2+z^2}$, $w = \frac{1}{\sqrt{1-x^2-y^2-z^2}}$

$$w = \sin^{-1} \left(\frac{1}{x+y+z} \right)$$

$f(x, y)$ is continuous at a given point $P(x_0, y_0)$ if

(i) $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x, y)$ exists

and unique limit.

(ii) $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x, y) = f(x_0, y_0)$

If any one of (i) and (ii) or both are not satisfied then $f(x, y)$ is said to be discontinuous at $P(x_0, y_0)$

(a) If (i) is not satisfied, discontinuity is of first kind or removable discontinuity.

(b) If (ii) is not satisfied, discontinuity is of second kind or (non-removable discontinuity)

① $f(x, y) = x^2 + y^2 + x$

$$f_x = \frac{\partial f}{\partial x} = D_x f = \frac{\partial}{\partial x} (x^2 + y^2 + x) \quad (\text{treat } y \text{ constant})$$
$$= 2x + 0 + 1$$
$$= 2x + 1$$

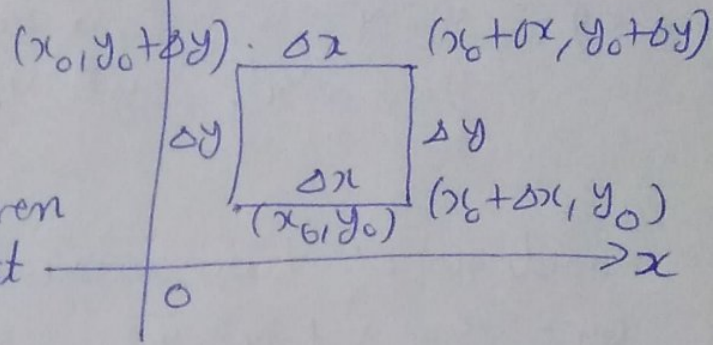
$$\text{and } f_y = \frac{\partial f}{\partial y} = D_y f = \frac{\partial}{\partial y} (x^2 + y^2 + x) \quad (\text{treat } x \text{ constant})$$
$$= 0 + 2y + 0$$
$$= 2y$$

By definition (the first principle),

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x}$$

$$\frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y+\Delta y) - f(x, y)}{\Delta y}$$

f_x and f_y exist at a given point $P(x, y)$ if limits exist and are unique.



② $f(x, y) = y e^{-x}$
 $f_x = y(-e^{-x}) = -y e^{-x} \quad \because \frac{d}{dx}(e^{ax}) = a e^{ax}$
 $f_y = 1 \cdot e^{-x} = e^{-x}$

By the first principle -

$$\begin{aligned} f_x &= \frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x, y) - f(x, y)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{y e^{-(x+\Delta x)} - y e^{-x}}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{y e^{-x} (e^{-\Delta x} - 1)}{\Delta x} \end{aligned}$$

$$= y e^{-x} \lim_{\Delta x \rightarrow 0} \left[\frac{1 - \frac{\Delta x}{1!} + \frac{(\Delta x)^2}{2!} - \dots - 1}{\Delta x} \right]$$

$$\left\{ \because e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right\}$$

$$= y e^{-x} \lim_{\Delta x \rightarrow 0} \left[-1 + \frac{\Delta x}{2!} - \dots \right]$$

$$= y e^{-x} (-1 + 0 - \dots) = -y e^{-x}$$

$$\begin{aligned} f_y &= \frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y+\Delta y) - f(x, y)}{\Delta y} \\ &= \lim_{\Delta y \rightarrow 0} \frac{(y+\Delta y) e^{-x} - y e^{-x}}{\Delta y} \\ &= \lim_{\Delta y \rightarrow 0} \frac{\Delta y e^{-x}}{\Delta y} = e^{-x} \end{aligned}$$

$$\textcircled{3} f(x, y) = \sin(2x + 3y)$$

$$f_x = \cos(2x + 3y)(2) = 2 \cos(2x + 3y)$$

$$f_y = \cos(2x + 3y)(3) = 3 \cos(2x + 3y)$$

$$\left\{ \because \frac{d}{dx} \sin(ax + b) = a \cos(ax + b) \right\}$$

By the first principle -

$$\begin{aligned} f_x = \frac{\partial f}{\partial x} &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{\sin(2(x + \Delta x) + 3y) - \sin(2x + 3y)}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{2 \cos \frac{2(x + \Delta x) + 3y + (2x + 3y)}{2} \sin \frac{2(x + \Delta x) + 3y - (2x + 3y)}{2}}{\Delta x} \end{aligned}$$

$$\left\{ \begin{aligned} \because \sin(A + B) &= \sin A \cos B + \cos A \sin B \\ \sin(A - B) &= \sin A \cos B - \cos A \sin B \end{aligned} \right.$$

$$\begin{aligned} \sin(A + B) - \sin(A - B) &= 2 \cos A \sin B \\ \sin C - \sin D &= 2 \cos \frac{C + D}{2} \sin \frac{C - D}{2} \end{aligned}$$

$$\begin{aligned} \sin(A + B) + \sin(A - B) &= 2 \sin A \cos B \\ \sin C + \sin D &= 2 \sin \frac{C + D}{2} \cos \frac{C - D}{2} \end{aligned}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{2 \cos(2x + 3y + \Delta x) \sin \Delta x}{\Delta x}$$

$$= 2 \cos(2x + 3y + 0) \lim_{\Delta x \rightarrow 0} \left(\frac{\sin \Delta x}{\Delta x} \right)$$

$$= 2 \cos(2x + 3y)(1) = 2 \cos(2x + 3y)$$

$$\left\{ \because \lim_{x \rightarrow 0} \frac{\sin(ax)}{ax} = 1 \right\}$$

$$f_y = \frac{\partial f}{\partial y} = \lim_{\Delta y \rightarrow 0} \frac{f(x, y + \Delta y) - f(x, y)}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\sin(2x + 3(y + \Delta y)) - \sin(2x + 3y)}{\Delta y}$$

$$\begin{aligned}
 &= \lim_{\Delta y \rightarrow 0} \frac{1}{\Delta y} \frac{2}{\Delta y} \cos(2x + 3(y + \Delta y)) + (2x + 3y) \\
 &\quad \sin(2x + 3(y + \Delta y)) - (2x + 3y) \\
 &= \lim_{\Delta y \rightarrow 0} \frac{1}{\Delta y} \frac{2}{\Delta y} \cos(2x + 3y + \frac{3\Delta y}{2}) \sin(\frac{3\Delta y}{2}) \\
 &= \lim_{\Delta y \rightarrow 0} 3 \cos(2x + 3y + \frac{3\Delta y}{2}) \frac{\sin(\frac{3\Delta y}{2})}{\frac{3\Delta y}{2}} \\
 &= 3 \cos(2x + 3y + 0) \lim_{\Delta y \rightarrow 0} \frac{\sin(\frac{3\Delta y}{2})}{\frac{3\Delta y}{2}} \\
 &= 3 \cos(2x + 3y) \cdot \left[\because \lim_{x \rightarrow 0} \frac{\sin(ax)}{ax} = 1 \right]
 \end{aligned}$$

④ Show that the function

$$f(x, y) = \begin{cases} (x+y) \sin\left(\frac{1}{x+y}\right) & x+y \neq 0 \\ 0 & x+y = 0 \end{cases}$$

is continuous at $(0, 0)$ but its partial derivatives f_x and f_y do not exist at $(0, 0)$.

$$\begin{aligned}
 \text{Since } \lim_{(x, y) \rightarrow (0, 0)} f(x, y) &= \lim_{(x, y) \rightarrow (0, 0)} (x+y) \sin\left(\frac{1}{x+y}\right) \\
 &= 0 \lim_{(x, y) \rightarrow (0, 0)} \sin\left(\frac{1}{x+y}\right) \\
 &= 0 \quad \left(\because -1 \leq \sin \theta \leq 1 \text{ for any } \theta \in \mathbb{R} \right) \\
 &= f(0, 0)
 \end{aligned}$$

$$\therefore \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0)$$

$\Rightarrow f(x, y)$ is continuous at $(0, 0)$.

$$\begin{aligned}
 f_x(0, 0) &= \lim_{\Delta x \rightarrow 0} \frac{f(0 + \Delta x, 0) - f(0, 0)}{\Delta x} \\
 &= \lim_{\Delta x \rightarrow 0} \frac{\Delta x \sin\left(\frac{1}{\Delta x}\right) - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \sin\left(\frac{1}{\Delta x}\right) \text{ which does not exist.}
 \end{aligned}$$

$$\begin{aligned}
 f_y(0, 0) &= \lim_{\Delta y \rightarrow 0} \frac{f(0, 0 + \Delta y) - f(0, 0)}{\Delta y} \\
 &= \lim_{\Delta y \rightarrow 0} \frac{\Delta y \sin\left(\frac{1}{\Delta y}\right) - 0}{\Delta y} = \lim_{\Delta y \rightarrow 0} \sin\left(\frac{1}{\Delta y}\right) \text{ which does not exist.}
 \end{aligned}$$

$\Rightarrow f_x$ and f_y do not exist at $(0,0)$.

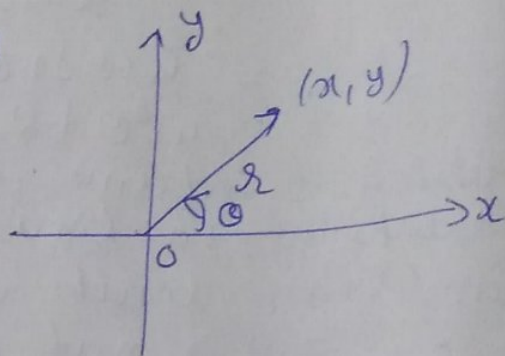
① Show that the function

$$f(x,y) = \begin{cases} \frac{x^2+y^2}{|x|+|y|} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

is continuous at $(0,0)$ but its partial derivatives f_x and f_y do not exist at $(0,0)$.

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{x^2+y^2}{|x|+|y|}$$

$\therefore$ let $x = r \cos \theta$ $y = r \sin \theta$
 $x^2 + y^2 = r^2$
 as $x, y \rightarrow 0$ $r \rightarrow 0$
 and θ is arbitrary
 $(0 \leq \theta \leq 2\pi)$



$$= \lim_{r \rightarrow 0} \frac{r^2 \cos^2 \theta + r^2 \sin^2 \theta}{r |\cos \theta| + r |\sin \theta|}$$

$$= \lim_{r \rightarrow 0} \frac{r}{|\cos \theta| + |\sin \theta|} = 0 = f(0,0)$$

$$\Rightarrow \lim_{(x,y) \rightarrow (0,0)} f(x,y) = f(0,0)$$

$\Rightarrow f(x,y)$ is continuous at $(0,0)$.

$$f_x(0,0) = \lim_{(\Delta x, 0) \rightarrow (0,0)} \frac{f(0+\Delta x, 0) - f(0,0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\frac{(\Delta x)^2}{|\Delta x|} - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{|\Delta x|} \text{ which does not exist.}$$

$$\therefore \lim_{x \rightarrow 0} \frac{x}{|x|} = \begin{cases} 1 & \text{when } x > 0 \\ -1 & \text{when } x < 0 \end{cases}$$

$$f_y(0,0) = \lim_{(0, \Delta y) \rightarrow (0,0)} \frac{f(0,0+\Delta y) - f(0,0)}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\frac{(\Delta y)^2}{|\Delta y|} - 0}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{\Delta y}{|\Delta y|} \text{ which does not exist.}$$

$$\therefore \lim_{y \rightarrow 0} \frac{y}{|y|} = \begin{cases} 1 & \text{when } y > 0 \\ -1 & \text{when } y < 0 \end{cases}$$

$\therefore f_x$ and f_y do not exist at $(0,0)$.

⑧ Show that the function $(x, y) \neq (1, 1)$

$$f(x, y) = \begin{cases} \frac{x^2 - y^2}{x - y} = x + y & (x, y) \neq (1, 1) \\ 2 & (x, y) = (1, 1) \end{cases}$$

is continuous and differentiable at $(1, 1)$.

Let $f(x, y) = 2$ $x + y = 1 + 1 = 2 = f(1, 1)$
 $(x, y) \rightarrow (1, 1)$ $(x, y) \rightarrow (1, 1)$

$\Rightarrow f(x, y)$ is continuous at $(1, 1)$.

$$f_x(1, 1) = \lim_{\Delta x \rightarrow 0} \frac{f(1 + \Delta x, 1) - f(1, 1)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{(1 + \Delta x) + 1 - 2}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} = 1$$

$$f_y(1, 1) = \lim_{\Delta y \rightarrow 0} \frac{f(1, 1 + \Delta y) - f(1, 1)}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{(1 + (1 + \Delta y)) - 2}{\Delta y}$$

$$= \lim_{\Delta y \rightarrow 0} \frac{\Delta y}{\Delta y} = 1$$

$$\Delta z = f(1 + \Delta x, 1 + \Delta y) - f(1, 1)$$

$$= ((1 + \Delta x) + (1 + \Delta y)) - 2$$

$$= \Delta x + \Delta y$$

$$dz = f_x(1, 1) \Delta x + f_y(1, 1) \Delta y = 1 \cdot \Delta x + 1 \cdot \Delta y$$

$$\therefore \Delta z - dz = (\Delta x + \Delta y) - (\Delta x + \Delta y)$$

$$\Rightarrow \lim_{\Delta p \rightarrow 0} \frac{\Delta z - dz}{\Delta p} = 0$$

Hence, $f(x, y)$ is differentiable at $(1, 1)$.

OR we can use the following result -

If the function $z = f(x, y)$ has continuous first order partial derivatives at a point $P(x, y)$ in D then $f(x, y)$ is differentiable at $P(x, y)$.



So we need to prove
 $\lim_{(x,y) \rightarrow (1,1)} f_x(x,y) = f_x(1,1)$ and $\lim_{(x,y) \rightarrow (1,1)} f_y(x,y) = f_y(1,1)$.

Now, we have -

$$f(x,y) = x + y$$

$$f_x = 1$$

$$f_y = 1$$

$$\therefore \lim_{(x,y) \rightarrow (1,1)} f_x(x,y) = 1 = f_x(1,1) \text{ and } \lim_{(x,y) \rightarrow (1,1)} f_y(x,y) = 1 = f_y(1,1)$$

$\Rightarrow f_x$ and f_y are continuous at $(1,1)$.

$\therefore$ using above result, $f(x,y)$ is differentiable at $(1,1)$.

Total differential -

for $f(x,y)$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \underline{\partial f}$$

for $f(x,y,z)$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

① Find the total differential of the following functions -

(i) $z = \tan^{-1}(x/y)$, $(x,y) \neq (0,0)$

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy$$

$$\frac{\partial z}{\partial x} = \frac{1}{1+(x/y)^2} \cdot \frac{\partial}{\partial x} \left(\frac{x}{y} \right) = \frac{y^2}{y^2+x^2} \left(\frac{1}{y} \right) = \frac{y}{x^2+y^2}$$

$$\frac{\partial z}{\partial y} = \frac{1}{1+(x/y)^2} \cdot \frac{\partial}{\partial y} \left(\frac{x}{y} \right) = \frac{y^2}{y^2+x^2} \left(\frac{-x}{y^2} \right) = \frac{-x}{x^2+y^2}$$

$$\therefore dz = \frac{y dx - x dy}{x^2+y^2} = \underline{\underline{\frac{1}{x^2+y^2} (y dx - x dy)}}$$

$$(ii) u = \left(xz + \frac{x}{z}\right)^y ; z \neq 0$$

$$du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy + \frac{\partial u}{\partial z} dz$$

$$\frac{\partial u}{\partial x} = y \left(xz + \frac{x}{z}\right)^{y-1} \frac{\partial}{\partial x} \left(xz + \frac{x}{z}\right)$$

$$= y \left(xz + \frac{x}{z}\right)^{y-1} \left(z + \frac{1}{z}\right)$$

$$\frac{\partial u}{\partial y} = \left(xz + \frac{x}{z}\right)^y \log_e \left(xz + \frac{x}{z}\right) \quad \left[\because \frac{d}{dx} (a^x) = a^x \log_e a\right]$$

$$\frac{\partial u}{\partial z} = y \left(xz + \frac{x}{z}\right)^{y-1} \frac{\partial}{\partial z} \left(xz + \frac{x}{z}\right)$$

$$= y \left(xz + \frac{x}{z}\right)^{y-1} \left(x - \frac{x}{z^2}\right)$$

$$\therefore du = \left(xz + \frac{x}{z}\right)^{y-1} \left[y \left(z + \frac{1}{z}\right) dx + xy \left(1 - \frac{1}{z^2}\right) dy \right]$$

$$+ \left[\left(xz + \frac{x}{z}\right)^y \log_e \left(xz + \frac{x}{z}\right) \right] dz$$

Q. Using differentials, find an approximate value of

$$(i) f(4.1, 4.9), \text{ where } f(x, y) = \sqrt{x^3 + x^2 y}$$

$$\text{We have } \Delta f \approx df$$

$$f(x + \Delta x, y + \Delta y) - f(x, y) = f_x \Delta x + f_y \Delta y$$

$$f(x + \Delta x, y + \Delta y) = f(x, y) + f_x \Delta x + f_y \Delta y$$

$$f(4.1, 4.9) = f(4 + 0.1, 5 - 0.1) = f(4, 5) + f_x(4, 5)(0.1) + f_y(4, 5)(-0.1)$$

$$\text{Here } \Delta x = 0.1 \text{ and } \Delta y = -0.1$$

$$f(4, 5) = \sqrt{4^3 + 4^2 \cdot 5} = 12$$

$$f_x(x, y) = \frac{3x^2 + 2xy}{2\sqrt{x^3 + x^2 y}}$$

$$f_x(4, 5) = \frac{11}{3}$$

$$f_y(x, y) = \frac{x^2}{2\sqrt{x^3 + x^2 y}}$$

$$f_y(4, 5) = \frac{8}{3}$$

$$\therefore f(4.1, 4.9) = 12 + \frac{11}{3}(0.1) + \frac{8}{3}(-0.1) = 12.3$$

ii) $f(x, y) = x^2 y$, where $f(x, y) = x^2 y$
 $f(2, 3) = f(2+0, 3+0) = f(2, 3) + f_x(2, 3)(0-2) + f_y(2, 3)(0-2)$
 $f(2, 3) = 2^3 = 8$
 $f_x(x, y) = y x^{y-1}$ $f_x(2, 3) = 12$
 $f_y(x, y) = x^2 \log x$ $f_y(2, 3) = 8(0-3 \log 2) = -24 \log 2$
 $\therefore f(2, 3) = 8 + 12(0-2) + (-24 \log 2)(0-2)$
 $= 8 + 12(-2) + 48 \log 2 = 8 - 24 + 48 \log 2 = 8 - 24 + 33.1774 = 17.1774$

i) $f(x, y) = \cos x \sin y$

To find $\cos 44^\circ \sin 39^\circ$.

$$f(44^\circ, 39^\circ) = f(45^\circ - 1^\circ, 30^\circ + 9^\circ)$$

$$= f(45^\circ, 30^\circ) + f_x(45^\circ, 30^\circ) \left(-\frac{\pi}{180}\right) + f_y(45^\circ, 30^\circ) \left(\frac{2\pi}{180}\right)$$

Here $\Delta x = -1^\circ = -\frac{\pi}{180}$ and $\Delta y = 9^\circ = \frac{2\pi}{180}$

$$f(45^\circ, 30^\circ) = \cos 45^\circ \sin 30^\circ = \frac{1}{2\sqrt{2}}$$

$$f_x(x, y) = -\sin x \sin y$$
 $f_x(45^\circ, 30^\circ) = -\sin 45^\circ \sin 30^\circ = -\frac{1}{2\sqrt{2}}$

$$f_y(x, y) = \cos x \cos y$$
 $f_y(45^\circ, 30^\circ) = \cos 45^\circ \cos 30^\circ = \frac{\sqrt{3}}{2\sqrt{2}}$

$$\cos 44^\circ \sin 39^\circ = \frac{1}{2\sqrt{2}} - \frac{1}{2\sqrt{2}} \left(-\frac{\pi}{180}\right) + \frac{\sqrt{3}}{2\sqrt{2}} \left(\frac{2\pi}{180}\right)$$

$$= \frac{1}{2\sqrt{2}} \left(1 + \frac{\pi}{180} (2\sqrt{3} + 1)\right)$$

21) Find the percentage error in the computed area of an ellipse when an error of 2% is made in measuring the major and minor axes.

let area of ellipse $A = \pi ab$

Now $\frac{\Delta a}{a} = 0.02 = \frac{\Delta b}{b}$

$\therefore \Delta A = \frac{\partial A}{\partial a} \Delta a + \frac{\partial A}{\partial b} \Delta b$

$\frac{\Delta A}{A} = \frac{\partial A}{\partial a} \pi b \frac{\Delta a}{\pi ab} + \pi a \frac{\Delta b}{\pi ab}$

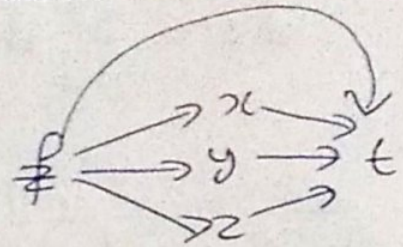
$= \frac{\Delta a}{a} + \frac{\Delta b}{b} = 0.02 + 0.02 = 0.04$

$\therefore \text{Percentage error} = \left| \frac{\Delta A}{A} \right| \times 100 = \underline{4\%}$

Derivation of Composite and Implicit functions (Chain Rule)-

$$\text{let } f = f(x, y, z)$$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

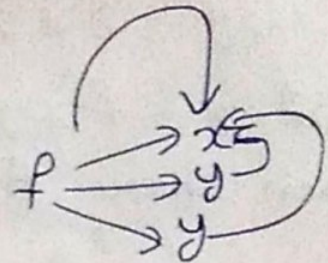


$$\text{let } x = x(t), y = y(t), z = z(t)$$

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}$$

$$\text{let } y = y(x), z = z(x)$$

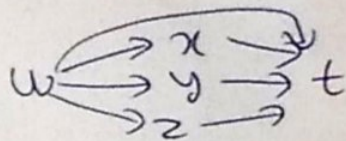
$$\frac{df}{dx} = \frac{\partial f}{\partial x} \frac{dx}{dx} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial z} \frac{dz}{dx}$$



$$\frac{df}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx} + \frac{\partial f}{\partial z} \frac{dz}{dx}$$

2) Find $\frac{dw}{dt}$ at $t=0$, where -

$$(w, y) = x \log y + y^x z \log z x; x = e^t, y = t e^t, z = t e^{-t}$$



$$dw = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$$

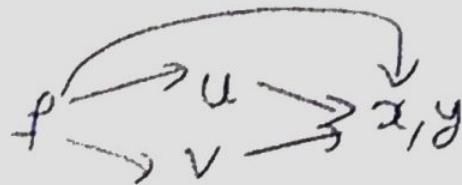
$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}$$

$$\begin{aligned} \frac{dw}{dt} &= (y+z) \left(\frac{dx}{dt} \right) + (x+z) (t e^t + e^t) + (y+x) (-t e^{-t} + e^{-t}) \\ &= 2(y+z)t + (x+z)(t+1)e^t + (x+y)(1-t)e^{-t} \end{aligned}$$

$$\therefore \frac{dw}{dt} \Big|_{t=0} = 0.$$

Let $z = f(u, v)$, $u = u(x, y)$, $v = v(x, y)$

$$\therefore dz = \frac{\partial f}{\partial u} du + \frac{\partial f}{\partial v} dv$$



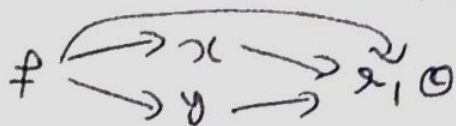
$$\frac{\partial z}{\partial x} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial x}$$

and $\frac{\partial z}{\partial y} = \frac{\partial f}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial f}{\partial v} \frac{\partial v}{\partial y}$

②. Let $z = f(x, y)$, $x = r \cos \theta$, $y = r \sin \theta$, then show that -

$$\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 = \left(\frac{\partial f}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial f}{\partial \theta}\right)^2$$

Here



$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial r}$$

$$\frac{\partial f}{\partial r} = \frac{\partial f}{\partial x} (\cos \theta) + \frac{\partial f}{\partial y} (\sin \theta) \longrightarrow \textcircled{1}$$

$$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta}$$

$$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} (-r \sin \theta) + \frac{\partial f}{\partial y} (r \cos \theta)$$

$$\frac{1}{r} \frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} (-\sin \theta) + \frac{\partial f}{\partial y} (\cos \theta) \longrightarrow \textcircled{2}$$

Squaring and adding ① and ② -

$$\left(\frac{\partial f}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial f}{\partial \theta}\right)^2 = \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 \quad \left\{ \because \cos^2 \theta + \sin^2 \theta = 1 \right\}$$

③ If $z = f(ax+by)$, then $\frac{\partial z}{\partial x} - a \frac{\partial z}{\partial y} = 0$.

$$\frac{\partial z}{\partial x} = f'(ax+by) a$$

$$\frac{\partial z}{\partial y} = f'(ax+by) b$$

$$\therefore b \frac{\partial z}{\partial x} - a \frac{\partial z}{\partial y} = f'(ax+by) ab - f'(ax+by) ab = 0.$$

④ If $z = \log\left(\frac{x^2-y^2}{x^2+y^2}\right)$, then $x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0$.

$$\begin{aligned} \frac{\partial z}{\partial x} &= \frac{1}{1 + \frac{x^2-y^2}{x^2+y^2}} \frac{\partial}{\partial x} \left(\frac{x^2-y^2}{x^2+y^2} \right) \\ &= \frac{x^2+y^2}{2x^2} \frac{(x^2+y^2)(2x) - (x^2-y^2)(2x)}{(x^2+y^2)^2} \\ &= \frac{2y^2}{x(x^2+y^2)} \end{aligned}$$

$$\begin{aligned} \frac{\partial z}{\partial y} &= \frac{1}{1 + \frac{x^2-y^2}{x^2+y^2}} \frac{\partial}{\partial y} \left(\frac{x^2-y^2}{x^2+y^2} \right) \\ &= \frac{x^2+y^2}{2x^2} \frac{(x^2+y^2)(-2y) - (x^2-y^2)(2y)}{(x^2+y^2)^2} \\ &= \frac{-2y}{x^2+y^2} \end{aligned}$$

$$\therefore x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{2y^2}{(x^2+y^2)} - \frac{2y^2}{(x^2+y^2)} = 0$$

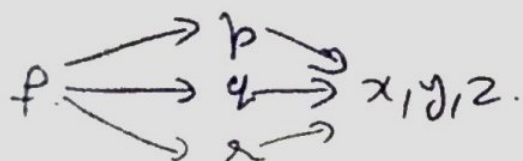
$$\Rightarrow x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 0.$$

Note- we will solve this example using Euler's Theorem for homogeneous functions. also.

⑤ If $u = f(x-y, y-z, z-x)$, then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.

Let $x-y = p$, $y-z = q$, $z-x = r$

$\therefore u = f(p, q, r)$; $p = x-y$, $q = y-z$, $r = z-x$.



$$\therefore df = \frac{\partial f}{\partial p} dp + \frac{\partial f}{\partial q} dq + \frac{\partial f}{\partial r} dr$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial x} + \frac{\partial f}{\partial r} \frac{\partial r}{\partial x} = \frac{\partial f}{\partial p} - \frac{\partial f}{\partial r}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial p} \frac{\partial p}{\partial y} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial y} + \frac{\partial f}{\partial r} \frac{\partial r}{\partial y} = -\frac{\partial f}{\partial p} + \frac{\partial f}{\partial q}$$

$$\frac{\partial f}{\partial z} = \frac{\partial f}{\partial p} \frac{\partial p}{\partial z} + \frac{\partial f}{\partial q} \frac{\partial q}{\partial z} + \frac{\partial f}{\partial r} \frac{\partial r}{\partial z} = -\frac{\partial f}{\partial q} + \frac{\partial f}{\partial r}$$

$$\therefore \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = 0$$

$$\Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$

⑥ If $z = f(x, y)$; $x = \lambda \cosh \theta$, $y = \lambda \sinh \theta$;
then

$$\left(\frac{\partial f}{\partial x}\right)^2 - \left(\frac{\partial f}{\partial y}\right)^2 = \left(\frac{\partial f}{\partial \lambda}\right)^2 - \frac{1}{\lambda^2} \left(\frac{\partial f}{\partial \theta}\right)^2$$

Now, we have -

$$\sinh \theta = \frac{e^\theta - e^{-\theta}}{2}$$

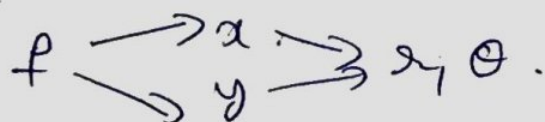
$$\cosh \theta = \frac{e^\theta + e^{-\theta}}{2}$$

$$\frac{d}{d\theta} (\sinh \theta) = \frac{e^\theta + e^{-\theta}}{2} = \cosh \theta$$

$$\frac{d}{d\theta} (\cosh \theta) = \frac{e^\theta - e^{-\theta}}{2} = \sinh \theta$$

Also $\cosh^2 \theta - \sinh^2 \theta = 1$.

Here



$$\therefore df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta} = \frac{\partial f}{\partial x} (\cosh \theta) + \frac{\partial f}{\partial y} (\sinh \theta) \quad \text{--- (1)}$$

$$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial \theta} = \frac{\partial f}{\partial x} (x \sinh \theta) + \frac{\partial f}{\partial y} (x \cosh \theta)$$

$$\frac{1}{x} \frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x} (\sinh \theta) + \frac{\partial f}{\partial y} (\cosh \theta) \quad \text{--- (2)}$$

Squaring and subtracting (1) and (2) -

$$\left(\frac{\partial f}{\partial x}\right)^2 - \frac{1}{x^2} \left(\frac{\partial f}{\partial \theta}\right)^2 = \left(\frac{\partial f}{\partial x}\right)^2 - \left(\frac{\partial f}{\partial y}\right)^2 \quad \left[\because \cosh^2 \theta - \sinh^2 \theta = 1 \right]$$

~~$$\textcircled{1} \frac{\partial f}{\partial z} = f(x, y); \quad x = e^{2u} + e^{-2v}, \quad y = e^{-2u} + e^{2v}$$~~

①

②

③

② $\partial f z = f(x, y); \quad x = e^{2u} + e^{-2v}; \quad y = e^{-2u} + e^{2v};$
then show that

$$\frac{\partial f}{\partial u} - \frac{\partial f}{\partial v} = 2 \left[x \frac{\partial f}{\partial x} - y \frac{\partial f}{\partial y} \right]$$

Here

$$\begin{matrix} & \nearrow x & \\ f & & \\ & \searrow y & \end{matrix} \rightarrow u, v.$$

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

$$\frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial u} = \frac{\partial f}{\partial x} (2e^{2u}) + \frac{\partial f}{\partial y} (-2e^{-2u})$$

$$\frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial f}{\partial y} \frac{\partial y}{\partial v} = \frac{\partial f}{\partial x} (-2e^{-2v}) + \frac{\partial f}{\partial y} (2e^{2v})$$

$$\therefore \frac{\partial f}{\partial u} - \frac{\partial f}{\partial v} = 2 \left[(e^{2u} + e^{-2v}) \frac{\partial f}{\partial x} - (e^{-2u} + e^{2v}) \frac{\partial f}{\partial y} \right]$$

$$\Rightarrow \frac{\partial f}{\partial u} - \frac{\partial f}{\partial v} = 2 \left[x \frac{\partial f}{\partial x} - y \frac{\partial f}{\partial y} \right]$$

Q. If $w = \sqrt{x^2 + y^2 + z^2}$; $x = u \cos v$, $y = u \sin v$, $z = uv$,

then

$$u \frac{\partial w}{\partial u} - v \frac{\partial w}{\partial v} = \frac{u}{\sqrt{1+v^2}}$$

Here $w = \sqrt{(u \cos v)^2 + (u \sin v)^2 + (uv)^2} = \sqrt{u^2 + u^2 v^2} = u \sqrt{1+v^2}$

$$\frac{\partial w}{\partial u} = 1 \cdot \sqrt{1+v^2} \quad \frac{\partial w}{\partial v} = u \cdot \frac{1}{2} \frac{2v}{\sqrt{1+v^2}} = \frac{uv}{\sqrt{1+v^2}}$$

$$\begin{aligned} \therefore u \frac{\partial w}{\partial u} - v \frac{\partial w}{\partial v} &= u \sqrt{1+v^2} - \frac{uv^2}{\sqrt{1+v^2}} \\ &= \frac{u(1+v^2) - uv^2}{\sqrt{1+v^2}} \\ &= \frac{u}{\sqrt{1+v^2}} \end{aligned}$$

Q. let $u = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$

$$\frac{\partial u}{\partial x} = -\frac{1}{2} \frac{2x}{(\sqrt{x^2 + y^2 + z^2})^{3/2}} = \frac{-x}{(\sqrt{x^2 + y^2 + z^2})^{3/2}}$$

$$\frac{\partial u}{\partial y} = \frac{-y}{(\sqrt{x^2 + y^2 + z^2})^{3/2}}$$

$$\frac{\partial u}{\partial z} = \frac{-z}{(\sqrt{x^2 + y^2 + z^2})^{3/2}}$$

$$\therefore \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 = \frac{x^2 + y^2 + z^2}{(x^2 + y^2 + z^2)^3} = \frac{1}{(x^2 + y^2 + z^2)^2}$$

Q. let $u = (x^2 + y^2 + z^2)^{n/2}$

$$\therefore \frac{\partial u}{\partial x} = \frac{n}{2} (x^2 + y^2 + z^2)^{\frac{n}{2}-1} (2x) = n(x^2 + y^2 + z^2)^{\frac{n}{2}-1} (x)$$

$$\frac{\partial u}{\partial y} = n(x^2 + y^2 + z^2)^{\frac{n}{2}-1} (y)$$

$$\frac{\partial u}{\partial z} = n(x^2 + y^2 + z^2)^{\frac{n}{2}-1} (z)$$

$$\begin{aligned} \left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial u}{\partial z}\right)^2 &= n^2 (x^2 + y^2 + z^2)^{n-2} (x^2 + y^2 + z^2) \\ &= n^2 (x^2 + y^2 + z^2)^{n-1} \end{aligned}$$